Spatial-Temporal GNNs

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  - GCN
  - TCN
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  - ST-GCN
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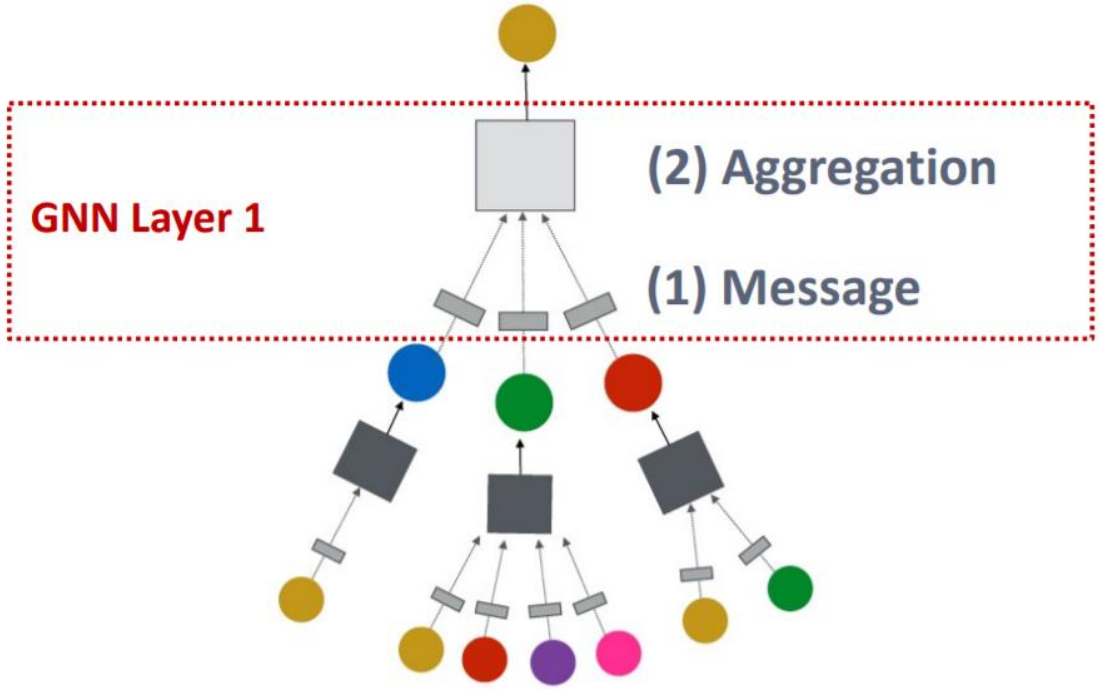
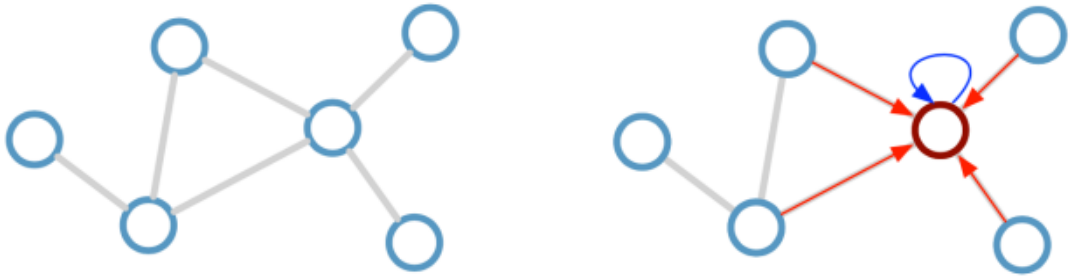
Recap: Graph Convolutional Networks (GCNs)

- GNN Layer = **Message + Aggregation**
  - Message COMPUTATION
    - How to make each neighbourhood node as embedding?
  - Message AGGERGATION
    - How to combine those embeddings?





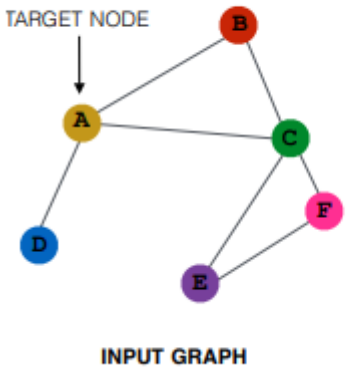
Update rule: $$\mathbf{h}_i^{(l+1)} = \sigma\left(\mathbf{h}_i^{(l)}\mathbf{W}_0^{(l)} + \sum_{j\in\mathcal{N}_i}\frac{1}{c_{ij}}\mathbf{h}_j^{(l)}\mathbf{W}_1^{(l)}\right)$$

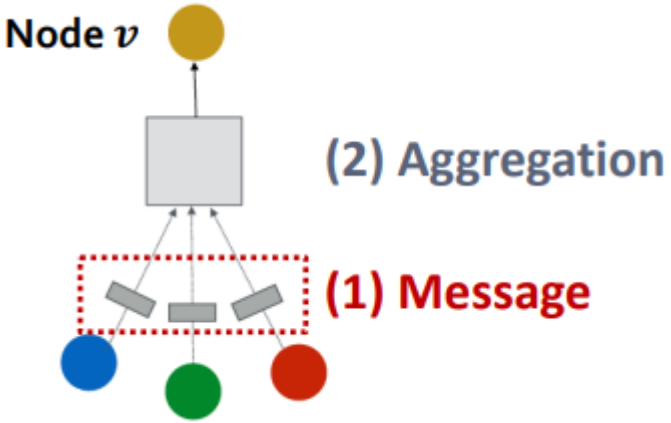
Kipf & Welling (ICLR 2017), related previous works by Duvenaud et al. (NIPS 2015) and Li et al. (ICLR 2016) c

GNN Layer: Message Computation

- Intuition: Each node will create a message, which will be sent to other nodes later
- Example: A Linear layer $\mathbf{m}_u^{(l)} = \mathbf{W}^{(l)}\mathbf{h}_u^{(l-1)}$
  - Multiply node features with weight matrix $\mathbf{W}^{(l)}$

Message function: $\mathbf{m}_u^{(l)} = \mathrm{MSG}^{(l)}\left(\mathbf{h}_u^{(l-1)}\right)$





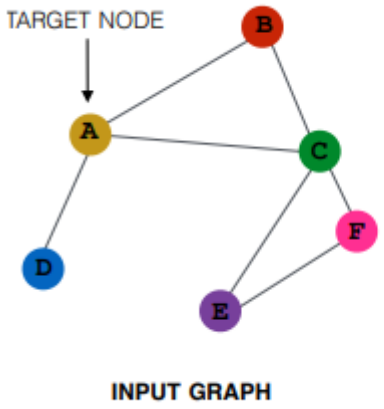
GNN Layer: Message Aggregation

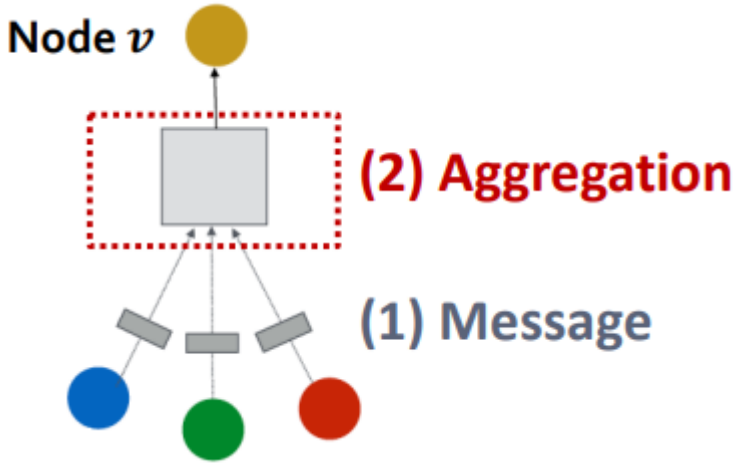
- Intuition: Each node will aggregate the messages from node v's neighbours

$$\mathbf{h}_v^{(l)} = \mathrm{AGG}^{(l)}\left(\left\{\mathbf{m}_u^{(l)}, u \in N(v)\right\}\right)$$

- Example: Sum(·), Mean(·) or Max(·) aggregator

$$\mathbf{h}_v^{(l)} = \mathrm{Sum}(\{\mathbf{m}_u^{(l)}, u \in N(v)\})$$





Recap: Temporal Convolutional Networks (TCN)

- To make human action recognition accurate and interpretable
- TCN avoid recurrence (RNNs/LSTMs) and use dilated temporal convolutions, which allow them to capture long-range dependencies

- Learn filters correspond to human-interpretable motion patterns:
  - Movement of specific joints
  - Consistent temporal dynamics
  - Localized motion segments

Interpretable 3D Human Action Analysis with Temporal Convolutional Networks, CVPR 2017

Recap: Temporal Convolutional Networks (TCN)

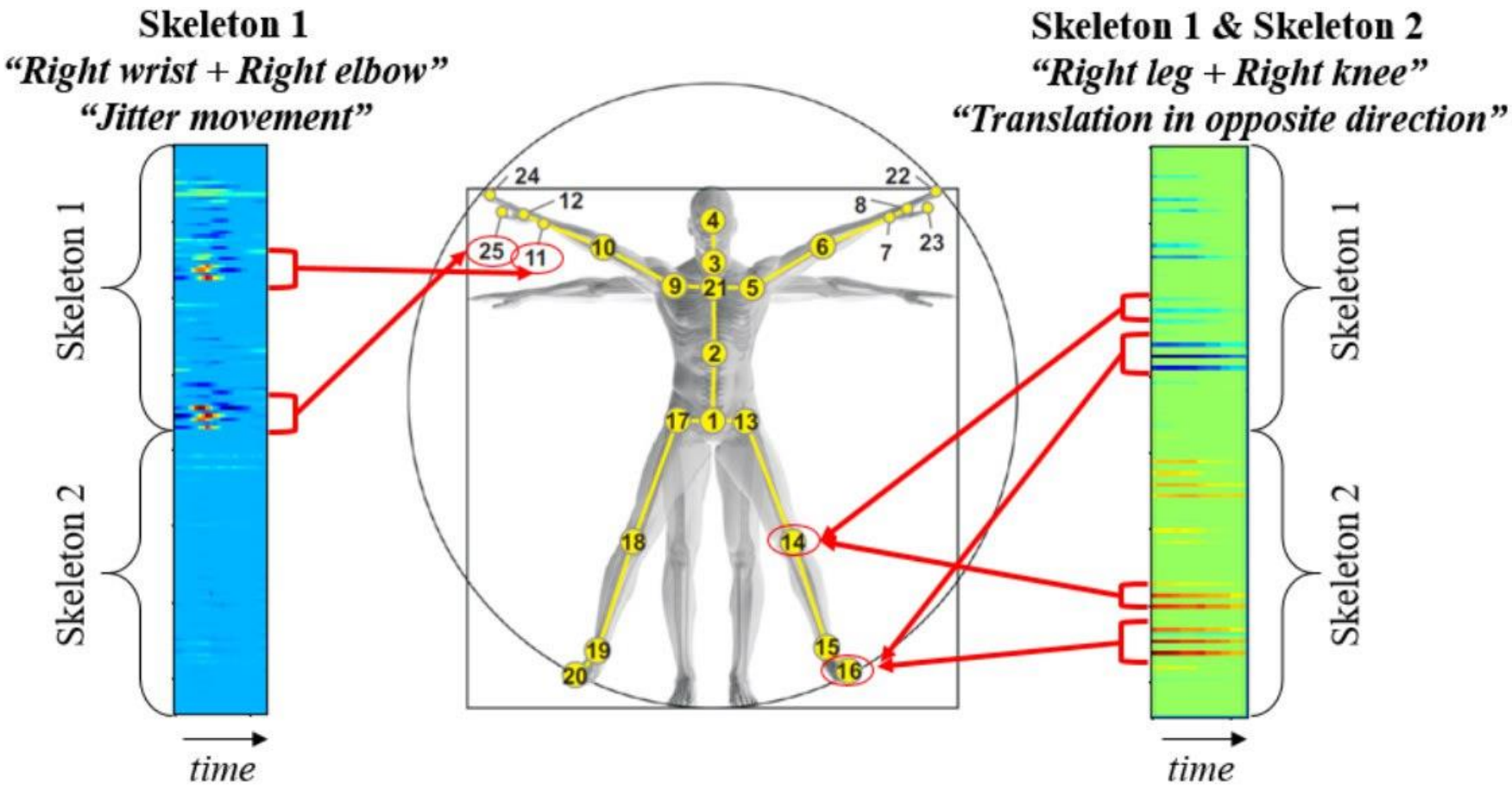
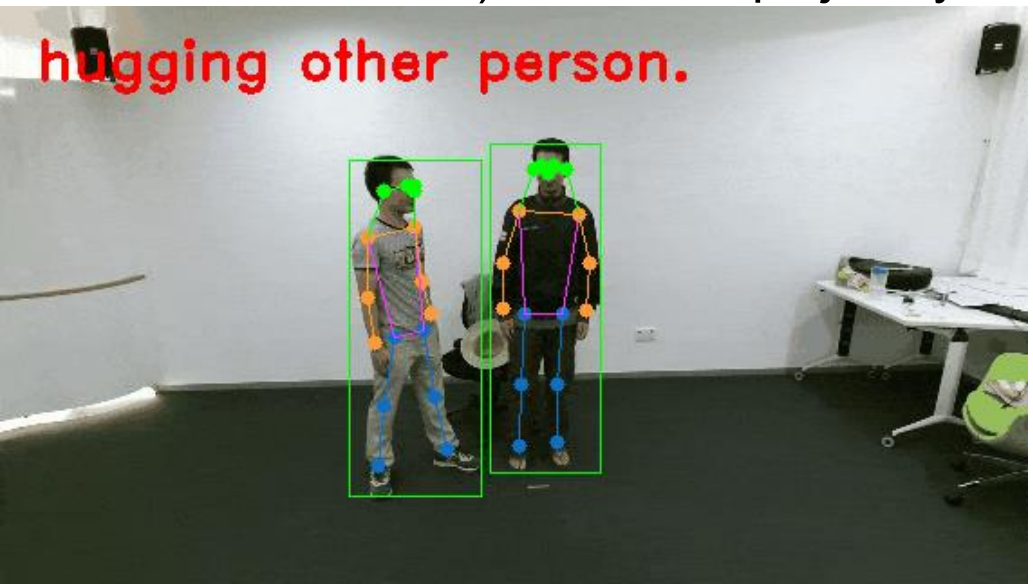


Figure 1: examples of direct mapping from layer 1 filter parameters to skeleton joints

Interpretable 3D Human Action Analysis with Temporal Convolutional Networks, CVPR 2017

Motivation

- Skeleton-based Action Recognition task
  - Input: video -> pose estimator -> skeleton sequence (2D/3D joint coordinates)
  - Skeleton is:
    - Compact & structured
    - Robust to appearance, background and lighting changes
    - More privacy than RGB videos
  - Target: classify actions such as hugging, dancing from sequences of skeletons
  - Challenge: need to model spatial (which joints move together?) and temporal (how they move over time?) relationships jointly



Motivation

- Each frame t:
  - Nodes: human joints (e.g. 25 joints)
  - Edges: physical bones (e.g., shoulder-elbow, elbow-wrist)
- Over time:
  - Connect the same joint across frames t, t+1, t+2,…
- Result: a spatial-temporal graph
  - Spatial edges: bones within a frame
  - Temporal edges: link across frames
- Run a GNN on this spatial-temporal graph

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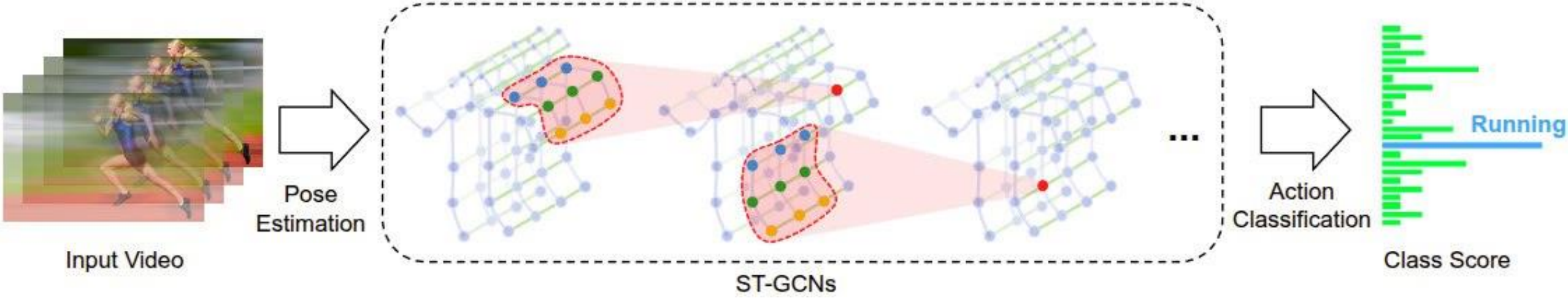
ST-GCN

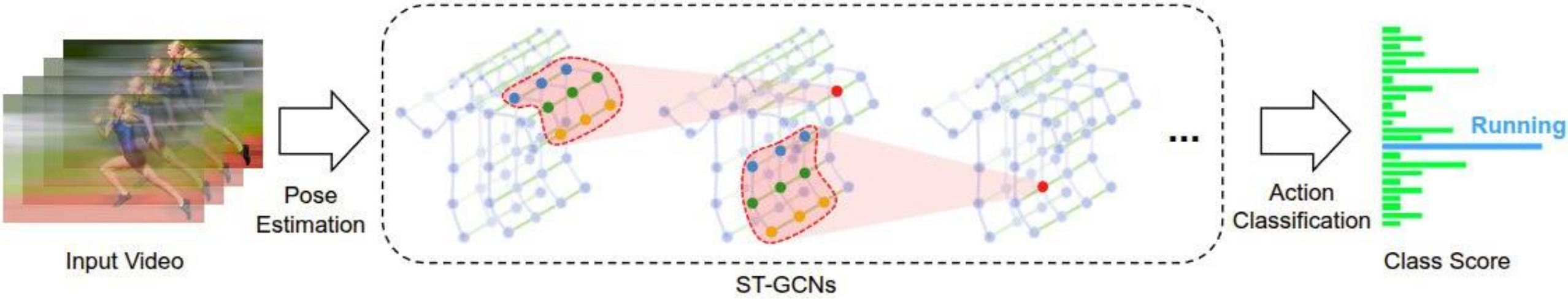
- Input tensor: $(N_{joints}, \mathrm{T}, C_{in})$
- Spatial GCN layer:
  - Partition adjacency into K matrices $\{A_k\}$
  - For each:
- Temporal Conv layer:
  - 1D convolution over time dimension (e.g., kernel size 9)
- Stack several ST-GCN blocks -> global pooling -> classifier



Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition, AAAI 2018

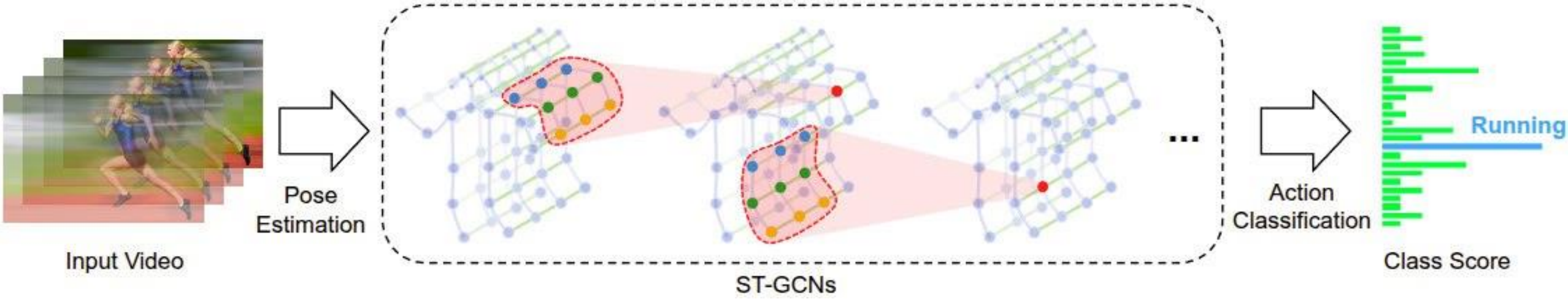
ST-GCN

- First model to treat skeletons as spatio-temporal graphs and apply GCN end-to-end
- Construct a spatial-temporal graph from skeleton sequence
  - Node: Human body joint
  - Edge:
    - Spatial edge: connect joints within a single frame
    - Temporal edge: connect the same joint across consecutive frames



Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition, AAAI 2018

- Pipeline:
  - Extract joint coordinates from the input video using a pose estimation algorithm
  - Construct a spatial-temporal graph from the extracted joint sequence
  - Pass the graph through multiple ST-GCN layers to progressively generate high-level feature maps
  - Classify the action using Softmax classifier



- Spatial graph convolution & partition strategies
  - Graph's neighbors do not have a fixed order like image pixels
  - Proposed to partition the neighbor set into several subsets and assign a single learnable weight to each subset
    - Uni-labeling: group all neighbors into a single set
    - Distance partitioning: partition neighbors based on their distance to the central node
    - Spatial configuration partitioning: partition neighbors into 3 subsets based on their distance to the skeleton's center of gravity:
      - Central node itself
      - Neighbors closer to the center of gravity (centripetal)
      - Neighbors farther from the center of gravity (centrifugal)

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- Learnable edge importance weighting
  - Not all joint edges are equally important for every action
  - Apply learnable mask to adjacency matrix
    - Ex:
    - Waving hand: wrist-elbow-shoulder connections are very important/ knee-ankle connection is not
    - Running: knee-ankle connection is more important
  - Learnable mask is created to allow the model to learn these differences by itself

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- Limitations:
  - Fixed graph topology:
  - Only use joint coordinates (first-order information)
  - Limited ability to capture long-range joint dependencies that are not directly connected by bones

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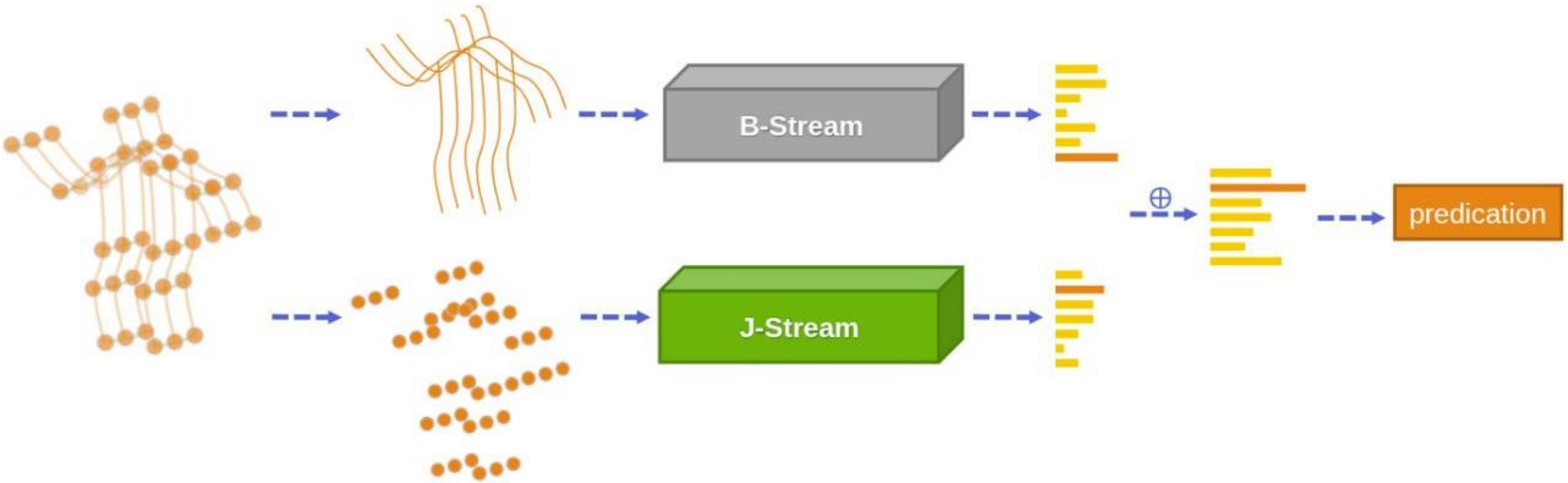
2s-AGCN

- Contributions:
  - Adaptive graph learning: learn adjacency, not fixed
  - Two streams: joint (position) stream + bone (motion) stream
- Model can adapt connections depending on action and person, while still leveraging prior skeletal structure



Two-Stream Adaptive Graph Convolutional Networks for Skeleton-Based Action Recognition, CVPR 2019

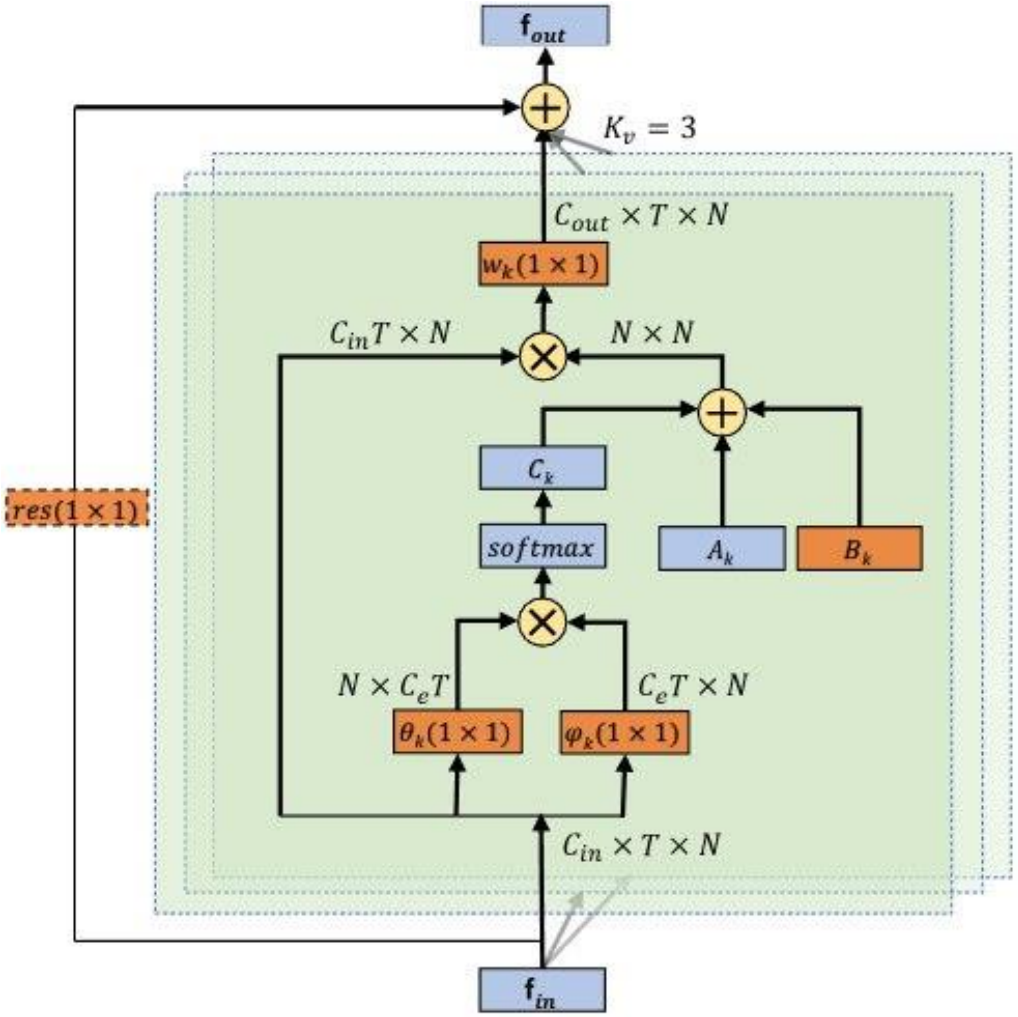
2s-AGCN

- Adaptive graph convolutional layer:
  - A_k: represent physical structure of human body
  - B_k: share learnable adjacency
  - C_k: data-dependent graph, learn unique graph for each sample

$$\mathbf{f}_{out} = \sum_{k}^{K_v} \mathbf{W}_k \mathbf{f}_{in} (\mathbf{A}_k + \mathbf{B}_k + \mathbf{C}_k)$$

where:

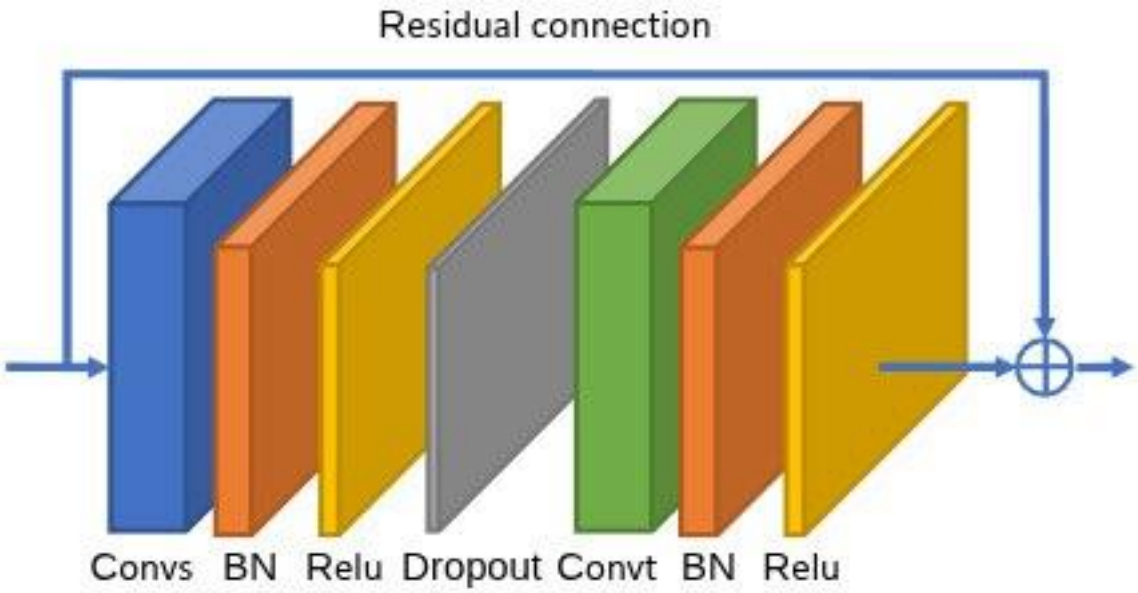
$$\mathbf{C}_k = softmax(\mathbf{f_{in}}^T \mathbf{W}_{\theta k}^T \mathbf{W}_{\phi k} \mathbf{f_{in}})$$



Two-Stream Adaptive Graph Convolutional Networks for Skeleton-Based Action Recognition, CVPR 2019

2s-AGCN

- Adaptive graph convolutional block:
  - Perform K_t x 1 convolution on CxTxN feature maps
  - One block is combination of one spatial GCN (Convs), one temporal GCN (Convt) with dropout layer and residual connection for stabilization



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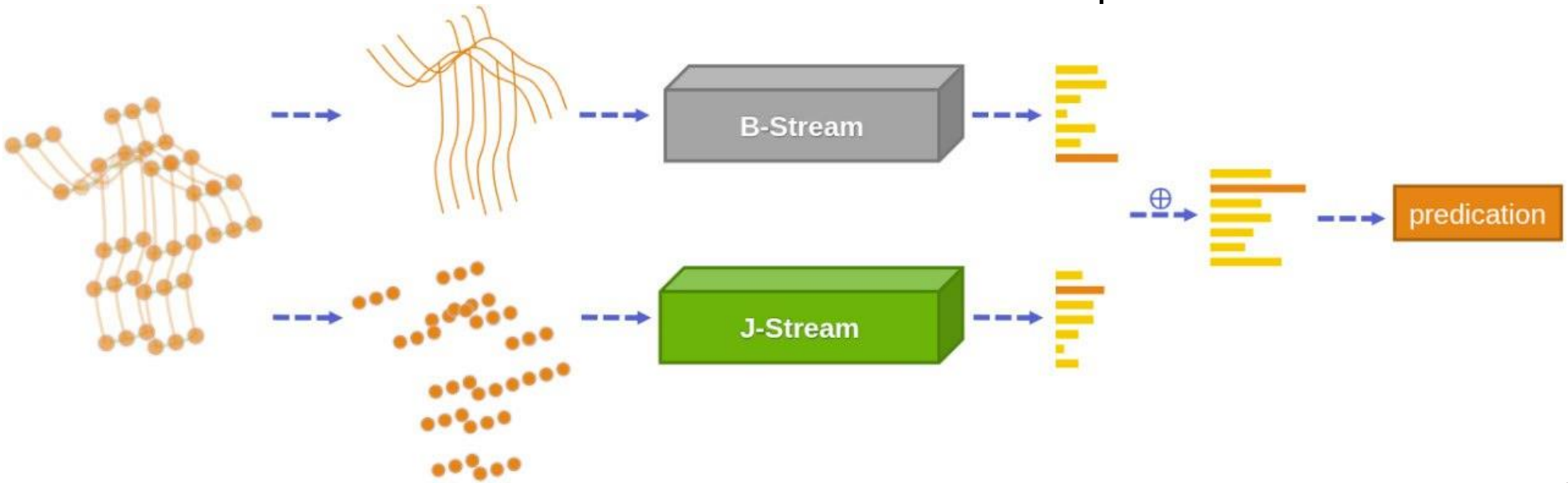
2s-AGCN

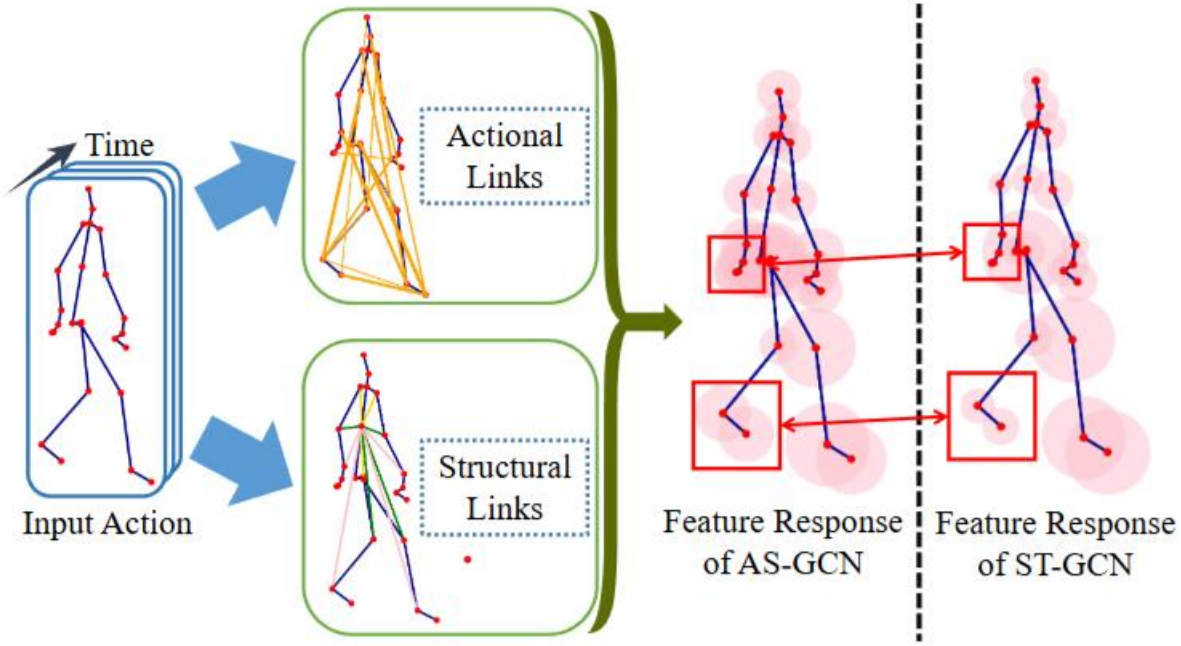
- Two-stream networks:
  - Bone information is second-order information
  - E.g.:
    - Source joint $v_1 = (x_1, y_1, z_1)$
    - Target joint $v_2 = (x_2, y_2, z_2)$
    - Bone $e_{v1,v2} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$
  - Joint data and bone data are fed into J-stream and B-stream
  - Softmax scores of 2 streams are added to obtain the fused score and predict action label



Two-Stream Adaptive Graph Convolutional Networks for Skeleton-Based Action Recognition, CVPR 2019

AS-GCN

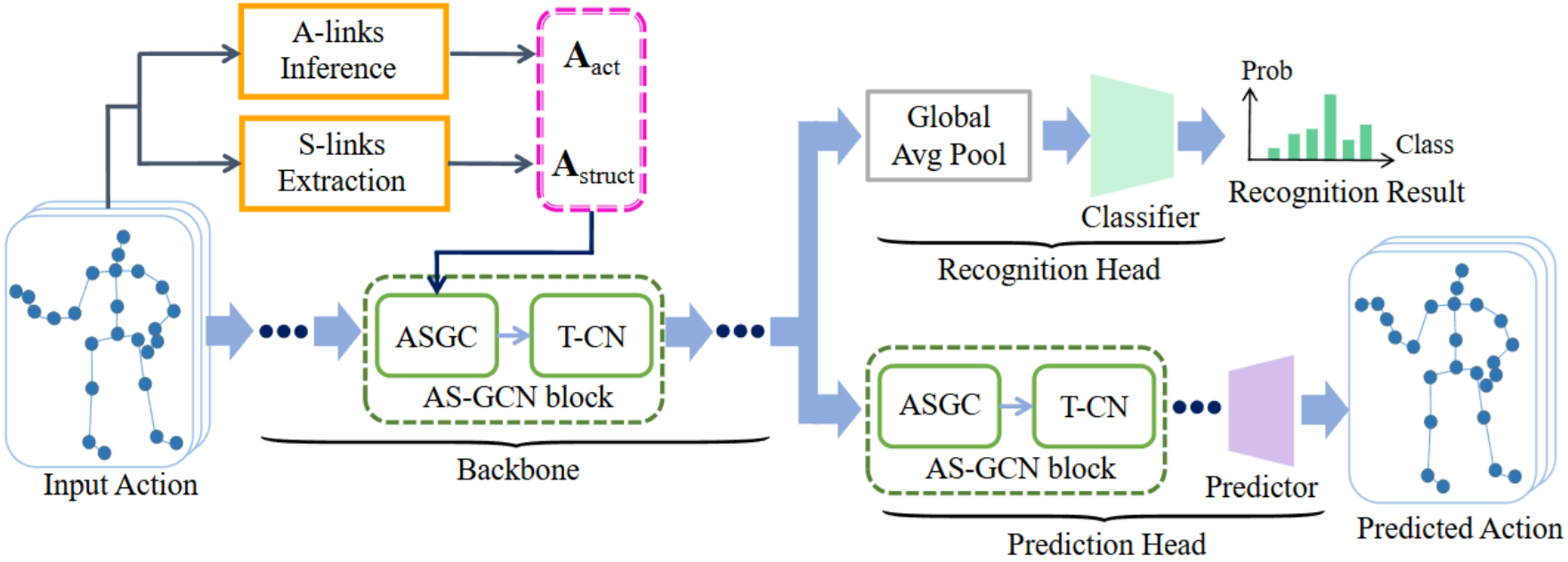
- Structural links: extended higher-order physical connections (beyond bones)
- Actional links: action-specific, latent dependencies between joints inferred from the sequence
- Architecture:
  - A-link inference module (encoder-decoder) to predict actional dependencies from observed motion
  - Combine actional + structural linkes -> generalized skeleton graph
- Add future pose prediction head as self-supervision, to help model capture finer dynamics



Actional-Structural Graph Convolutional Networks for Skeleton-based Action Recognition, CVPR 2019

AS-GCN

- Actional links (A-links)
  - Many actions need far-apart joints to move, leading to non-physical dependencies among joints
  - A-links are activated by actions and exist between arbitrary pair of joints



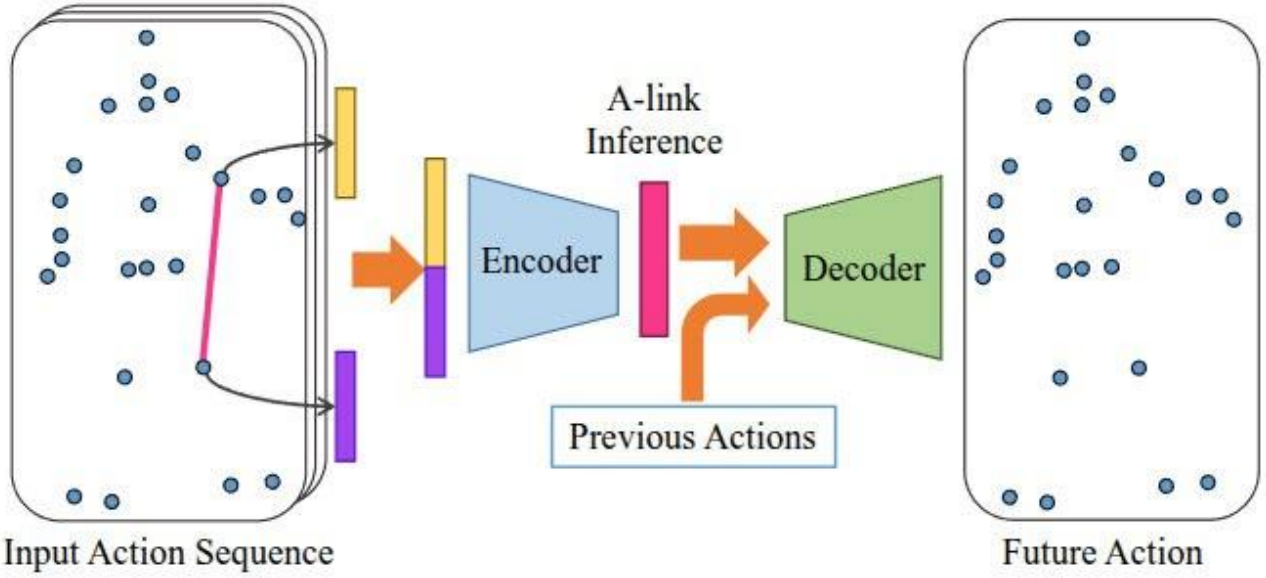
Actional-Structural Graph Convolutional Networks for Skeleton-based Action Recognition, CVPR 2019

- Actional links (A-links)
  - Many actions need far-apart joints, leading to non-physical dependencies among joints
  - A-links are activated by actions and exist between arbitrary pair of joints
  - Develop a trainable A-link inference module, which consist of encoder, decoder
- Encoder: estimate the states of A-links

$$\mathcal{A} = \text{encode}(\mathcal{X}) \in [0,1]^{n \times n \times C}$$

where: $A_{i,j,c}$ denote probability that i,j-th nodes are connected with c-th type

$\mathbf{x}_i = \text{vec}\left(\mathcal{X}_{i,:,:}\right) \in \mathbb{R}^{dT}$ is vector representation of i-th joint's feature across T frames



Propagate information between joints and links

$$\text{link features}: \mathbf{Q}_{i,j}^{(k+1)} = f_e^{(k)}(f_v^{(k)}(\mathbf{p}_i^{(k)}) \oplus (f_v^{(k)}(\mathbf{p}_j^{(k)})),$$

$$\text{joint features}: \mathbf{p}_i^{(k+1)} = \mathcal{F}(\mathbf{Q}_{i,:}^{(k+1)}) \oplus \mathbf{p}_i^{(k)},$$

After K times propagating, encoder outputs: $\mathcal{A}_{i,j,:} = \text{softmax}\left(\frac{\mathbf{Q}_{i,j}^{(K)} + \mathbf{r}}{\tau}\right) \in \mathbb{R}^C$

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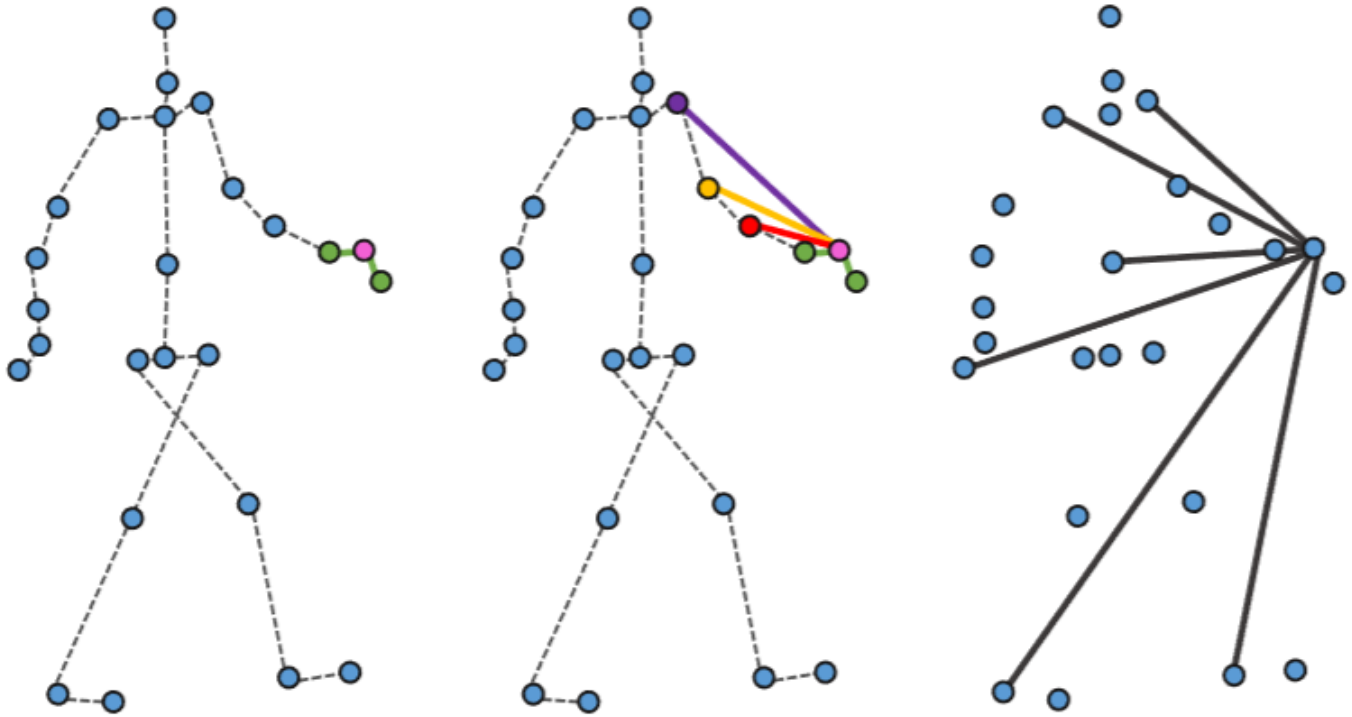
AS-GCN

- Actional-structural GCN
  - Action: walking
    - (a) skeleton links with limited neighboring range
    - (b) S-links let left hand to link to entire arm
    - (c) A-links capture long-range action-specific relations



(a) Skeleton graph (b) S-links (c) A-links

Actional-Structural Graph Convolutional Networks for Skeleton-based Action Recognition, CVPR 2019

AS-GCN

- Decoder: predict future 3D joint positions

$$\mathbf{X}_{t+1} = \text{decode}(\mathbf{X}_t, \ldots, \mathbf{X}_1, \mathcal{A}) \in \mathbb{R}^{n \times 3}$$

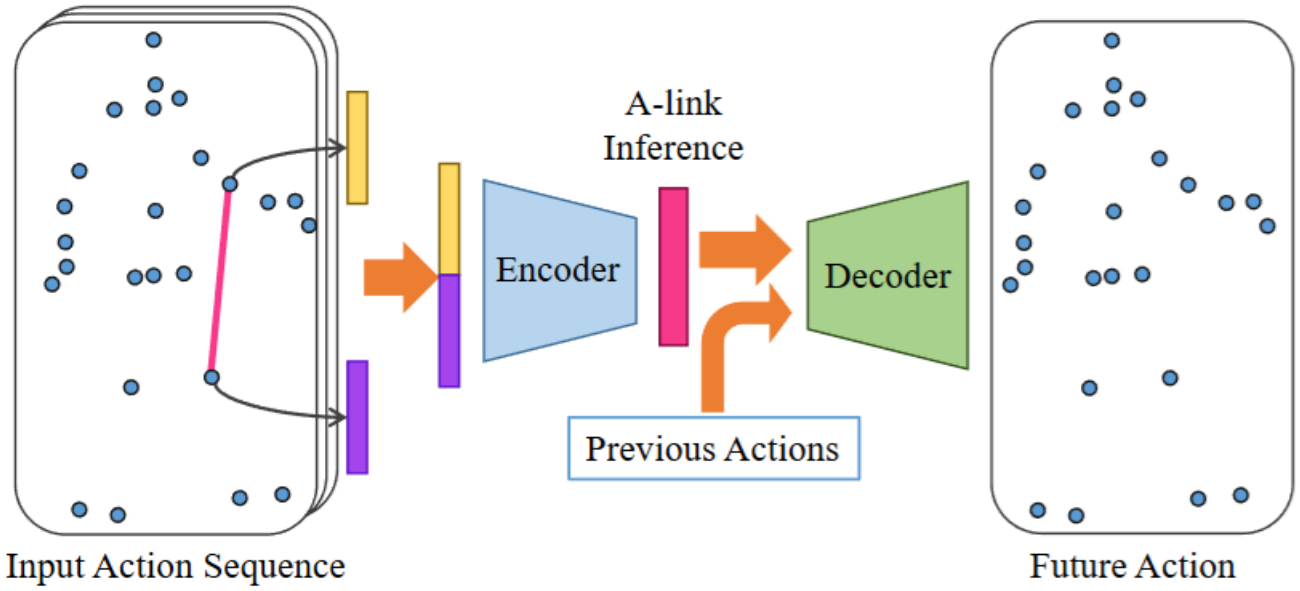
- Decode(.):

(a) $\mathbf{Q}_{i,j}^t = \sum_{c=1}^{C} \mathcal{A}_{i,j,c} f_e^{(c)}(f_v^{(c)}(\mathbf{x}_i^t) \oplus f_v^{(c)}(\mathbf{x}_j^t))$: generate link feature by weighted averaging on linking probabilities

(b) $\mathbf{p}_i^t = \mathcal{F}(\mathbf{Q}_{i,:}^t) \oplus \mathbf{x}_i^t$: aggregate link features to obtain corresponding joint features

(c) $\mathbf{S}_i^{t+1} = \text{GRU}(\mathbf{S}_i^t, \mathbf{p}_i^t)$: update joint features

(d) $\hat{\mu}_i^{t+1} = f_{\text{out}}(\mathbf{S}_i^{t+1}) \in \mathbb{R}^3$, : predict future joint positions



AS-GCN

- Structural links (S-links)
  - ST-GCN only aggregate 1-hop neighbors' information
  - Define structural graph convolution (SGC), which can directly reach L-hop neighbors to increase receptive field

$$
\begin{aligned}
\mathbf{X}_{\text{struc}} &= \text{SGC}\left(\mathbf{X}_{\text{in}}\right) \\
&= \sum_{l=1}^{L} \sum_{p \in \mathcal{P}} \mathbf{M}_{\text{struc}}^{(p,l)} \circ \hat{\mathbf{A}}^{(p)l} \mathbf{X}_{\text{in}} \mathbf{W}_{\text{struc}}^{(p,l)} \\
&\in \mathbb{R}^{n \times d_{\text{out}}},
\end{aligned}
$$

Where l is polynomial order,

$\hat{\mathbf{A}}^{(p)}$ is graph transition matrix,

$\mathbf{M}_{\text{struc}}^{(p,l)}, \mathbf{W}_{\text{struc}}^{(p,l)}$ are trainable weights to capture edge weights and feature importance

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AS-GCN

- Actional-structural graph convolution block:
  - Combine AGC and SGC and develop actional-structural graph convolution (ASGC)

$$\begin{aligned}\mathbf{X}_{\text{out}} &= \text{ASGC}\left(\mathbf{X}_{\text{in}}\right) \\ &= \mathbf{X}_{\text{struc}} + \lambda \mathbf{X}_{\text{act}} \in \mathbb{R}^{n \times d_{\text{out}}}\end{aligned}$$

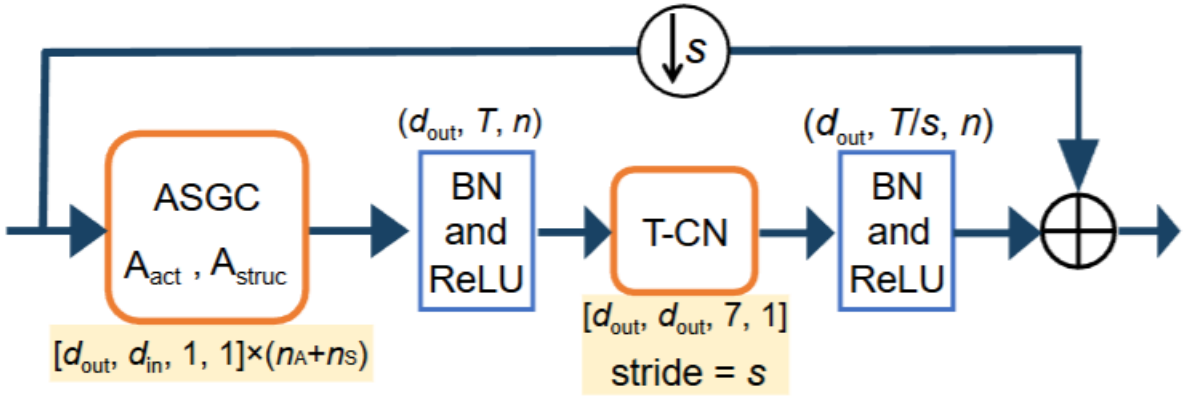
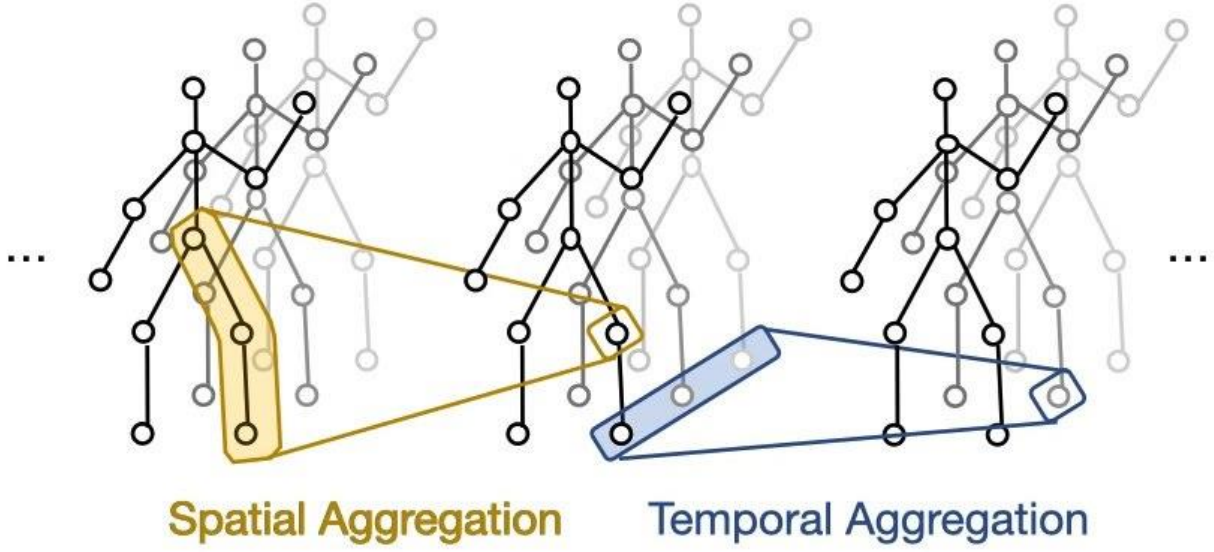


Figure : AS-GCN block

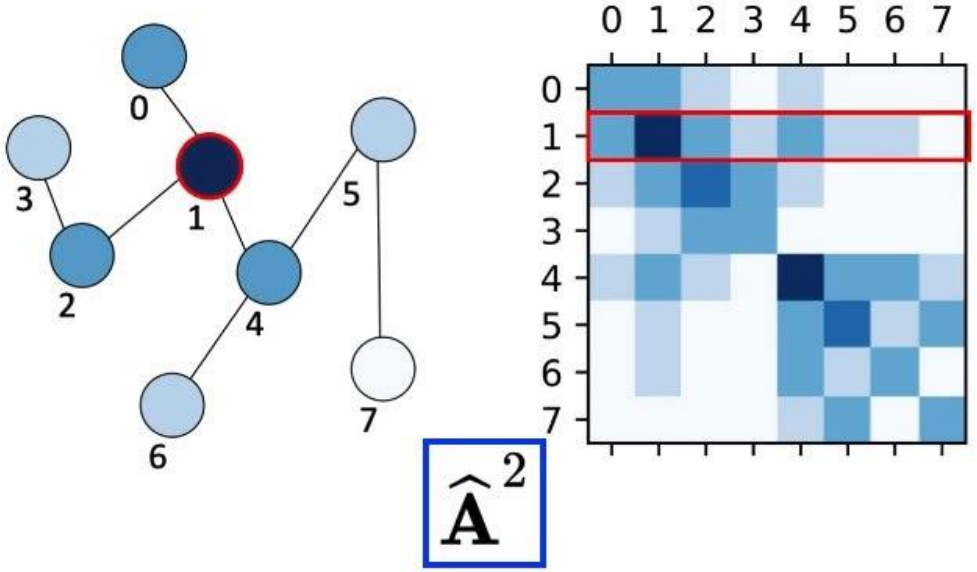
MS-G3D

- Existing graph-based approaches
- Factorized modeling:
  - GCNs + Temporal models



- Multi-scale graph convolutions:
  - GCNs over different adjacency powers

$$\mathbf{X}_t^{(l+1)} = \sigma\left(\sum_{k=0}^{K} \boxed{\widehat{\mathbf{A}}^k} \mathbf{X}_t^{(l)} \Theta_{(k)}^{(l)}\right)$$



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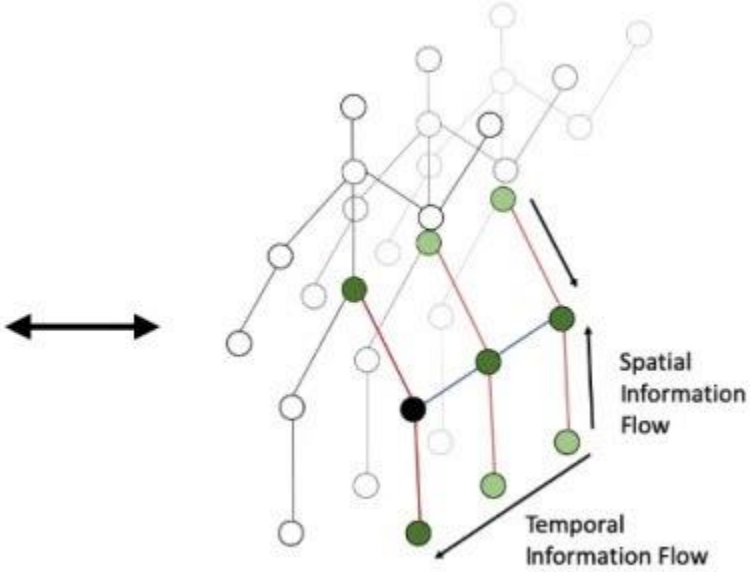
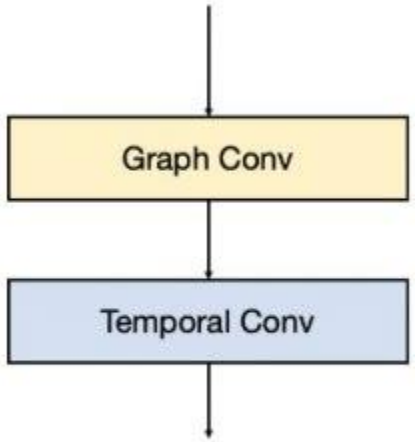
MS-G3D

- Factorized modeling:
  - Learn spatial-temporal features with spatial/temporal modules
  - Spatial: neighborhood aggregation (GCNs)
  - Temporal: node-wise sequence models (1D conv, TCN)





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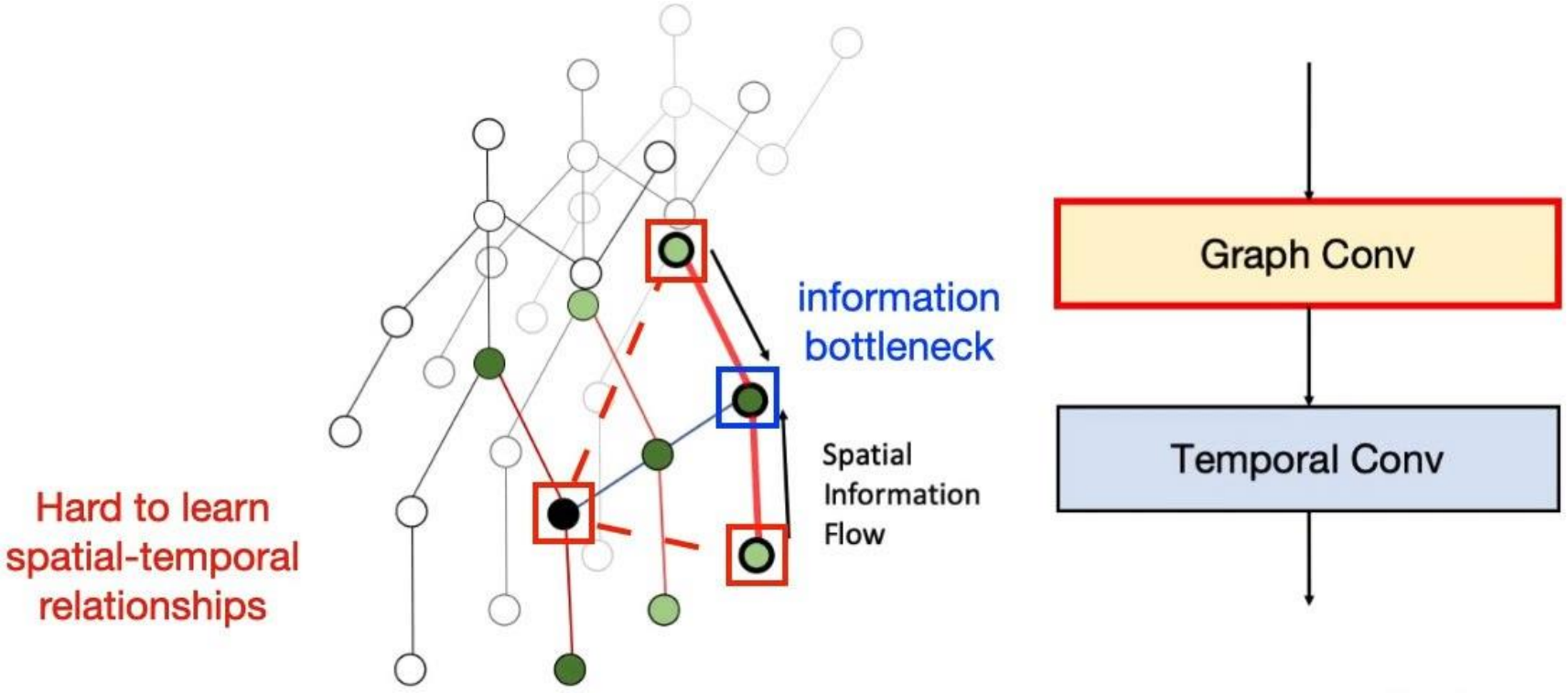
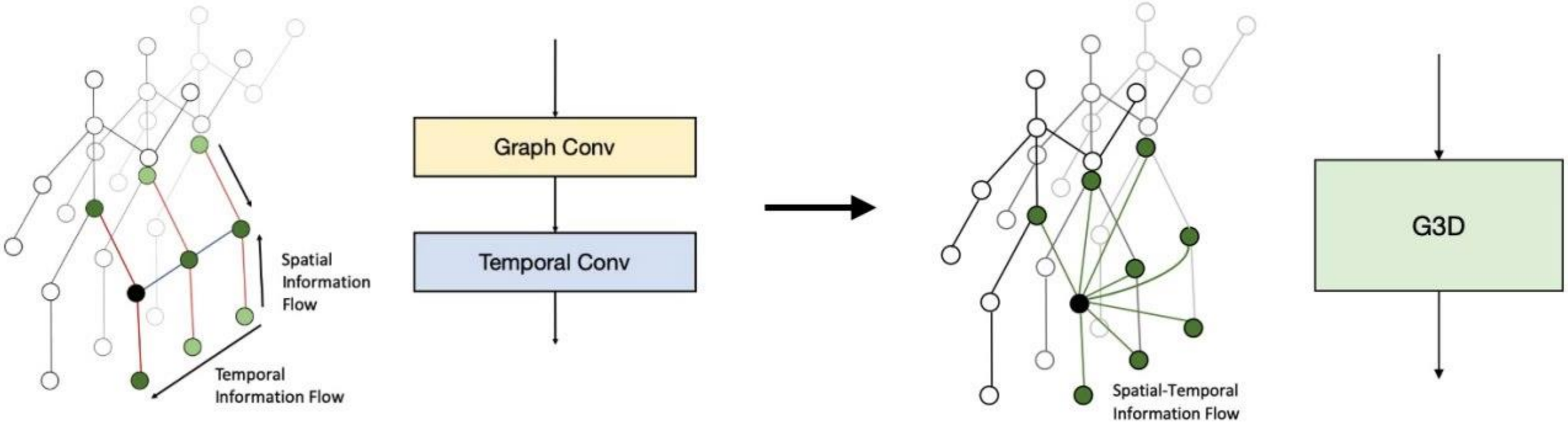
MS-G3D

- Factorized modeling:
  - Are indirect information flow
    - Factorization can create bottlenecks for feature propagation
    - Unweighted message passings (GCNs) can make aggregated feature generic



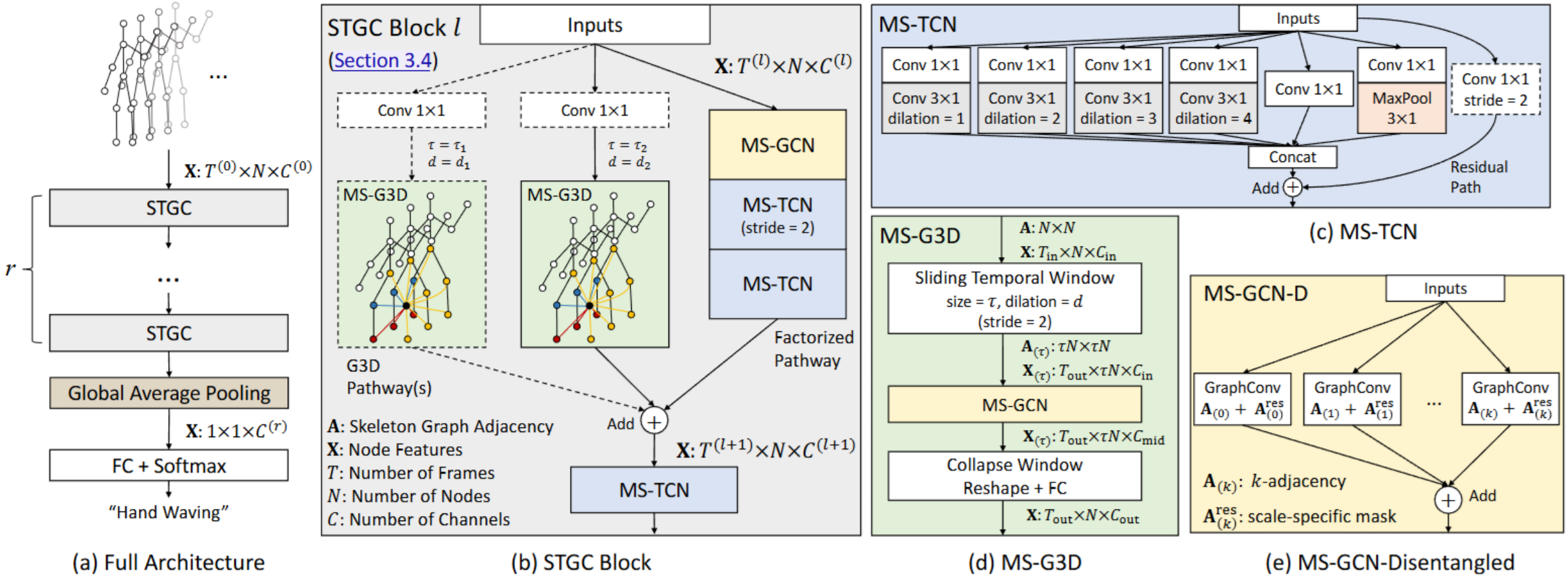
MS-G3D

- Apply unified spatial-temporal modeling
  - G3D modules: neighborhood aggregation across space and time
  - Edges serve as skip connections, allowing more direct information flow



MS-G3D

➢ Overall architecture:



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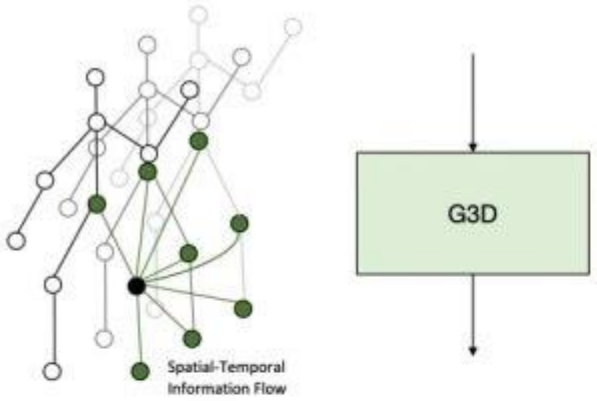
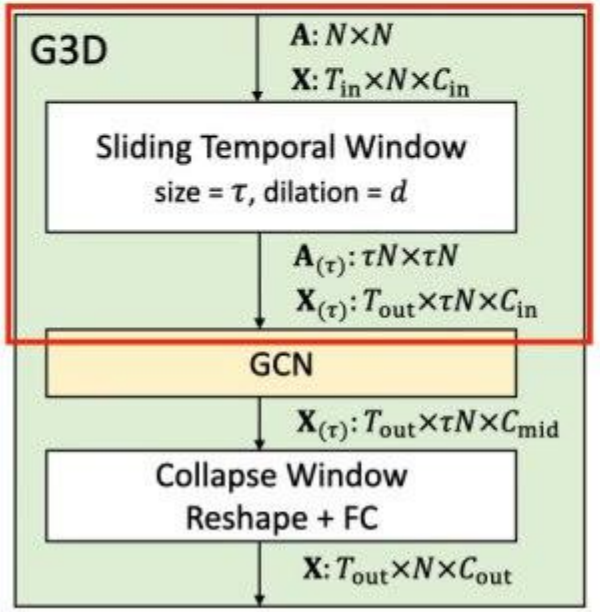
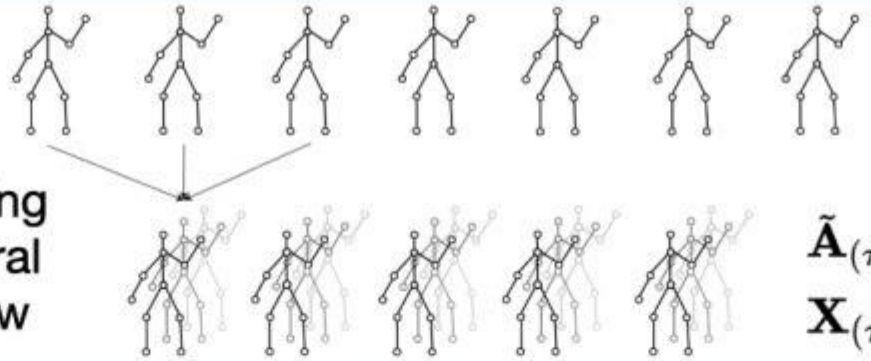
MS-G3D

- Unified spatial-temporal modeling







$\tilde{\mathbf{A}} \in \mathbb{R}^{N \times N}$ Spatial Graph

$\mathbf{X} \in \mathbb{R}^{T \times N \times C}$ Skeleton Features

(1) Sliding Temporal Window

$\tilde{\mathbf{A}}_{(\tau)} \in \mathbb{R}^{\tau N \times \tau N}$ Spatial-Temporal Window

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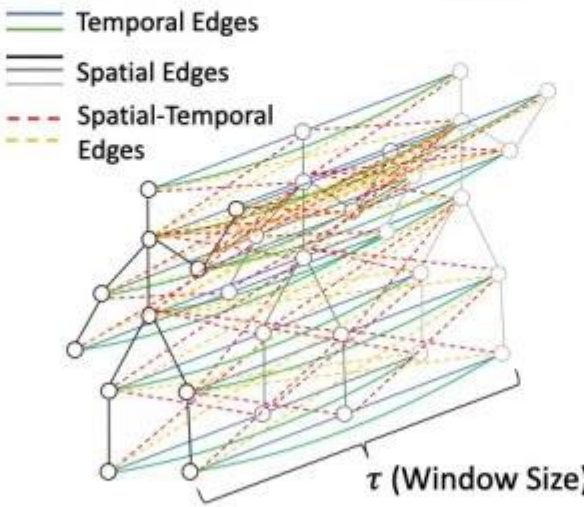

(2) **Extrapolate** Spatial Connectivity

$$\tilde{\mathbf{A}}_{(\tau)} = \begin{bmatrix} \tilde{\mathbf{A}} & \cdots & \tilde{\mathbf{A}} \\ \vdots & \ddots & \vdots \\ \tilde{\mathbf{A}} & \cdots & \tilde{\mathbf{A}} \end{bmatrix} \in \mathbb{R}^{\tau N \times \tau N}$$

(3) Graph Convolutions over **Windows**

$$[\mathbf{X}_{(\tau)}^{(l+1)}]_t = \sigma\left(\tilde{\mathbf{D}}_{(\tau)}^{-\frac{1}{2}} \tilde{\mathbf{A}}_{(\tau)} \tilde{\mathbf{D}}_{(\tau)}^{-\frac{1}{2}} [\mathbf{X}_{(\tau)}^{(l)}]_t \Theta^{(l)}\right)$$

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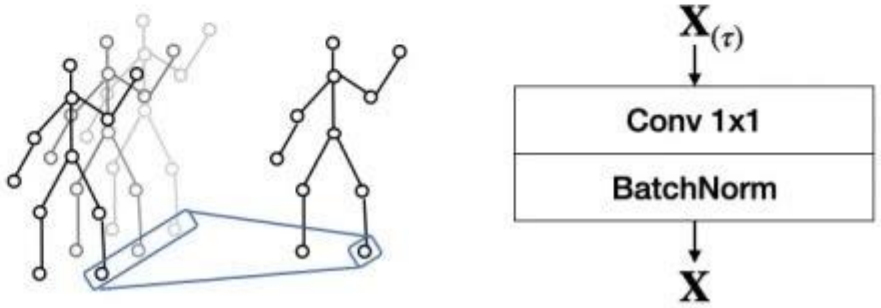
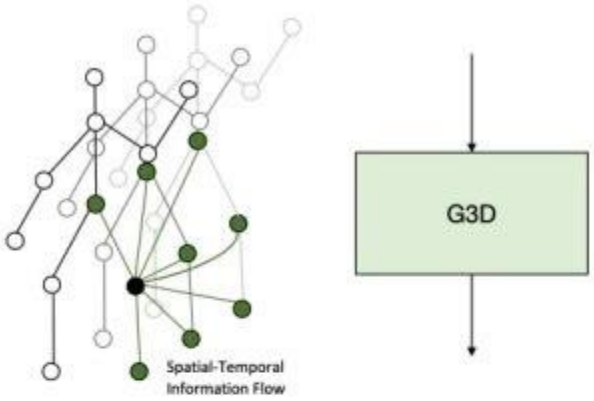
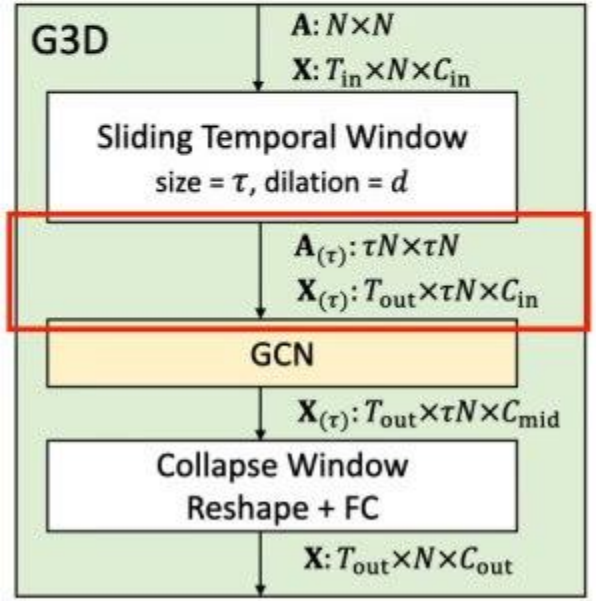
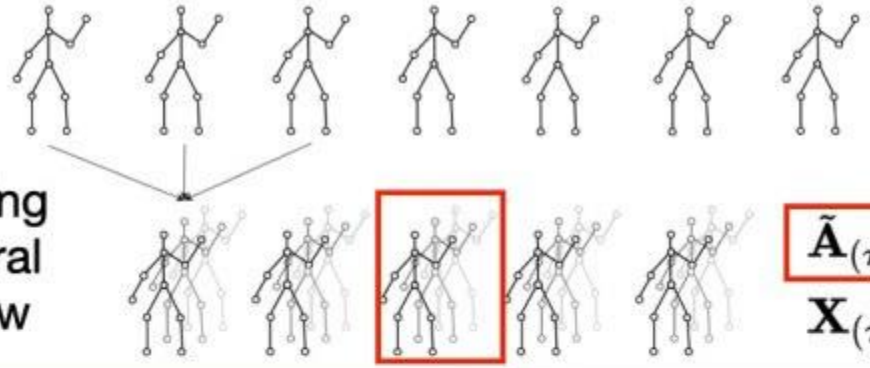


MS-G3D

- Unified spatial-temporal modeling







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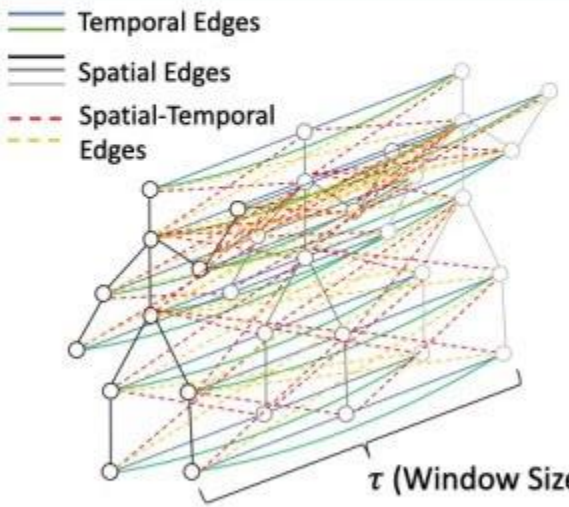

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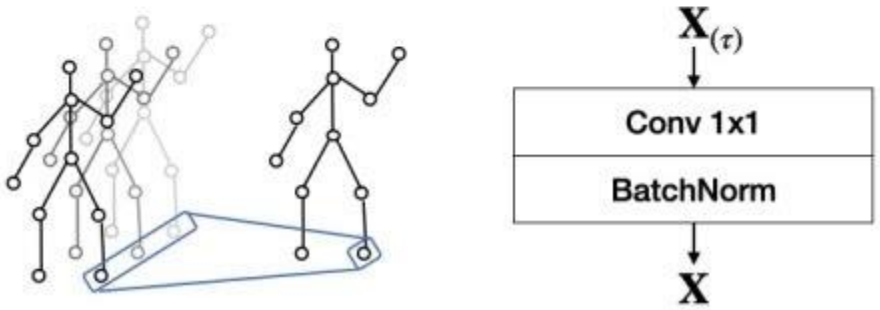
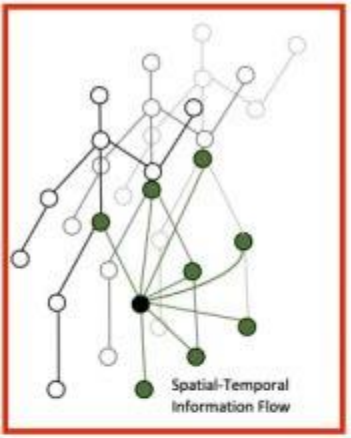
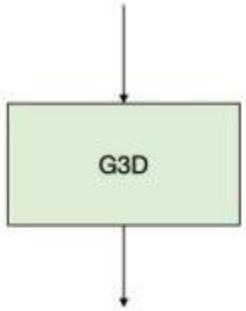
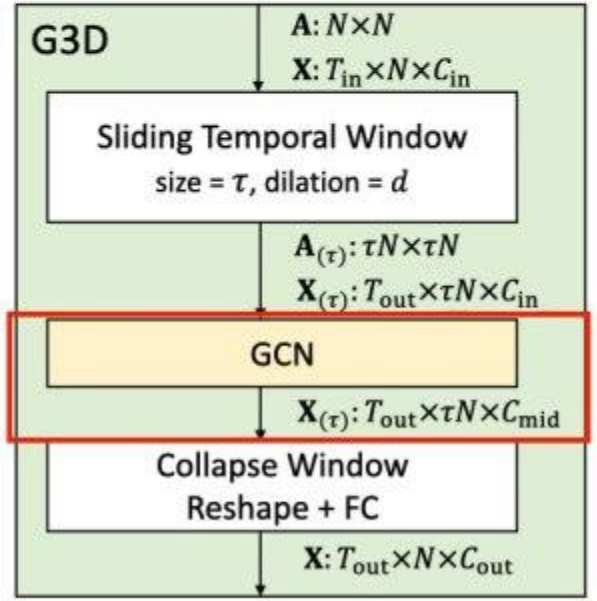


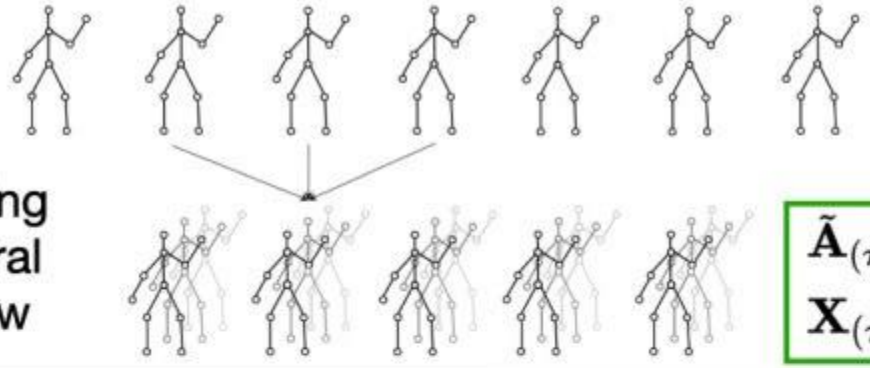
MS-G3D

- Unified spatial-temporal modeling









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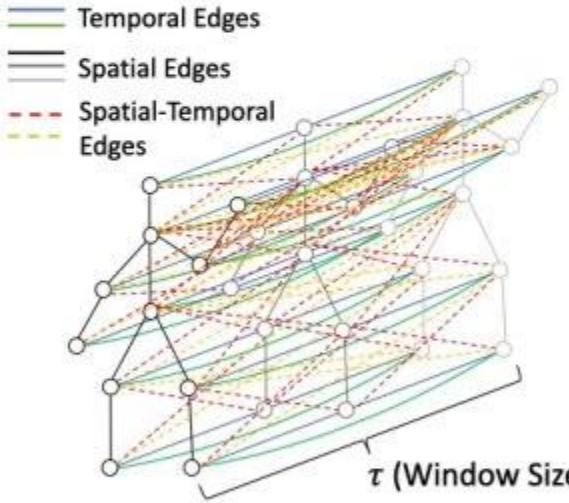

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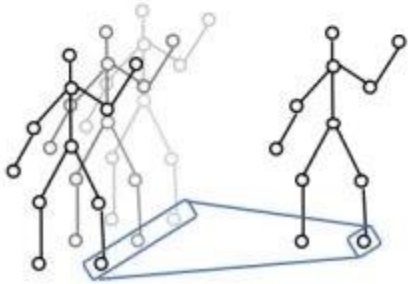
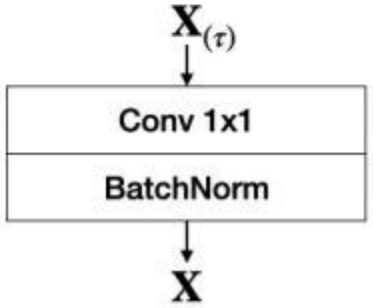
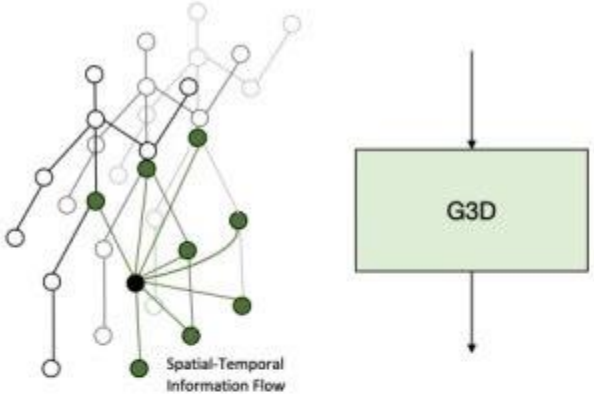
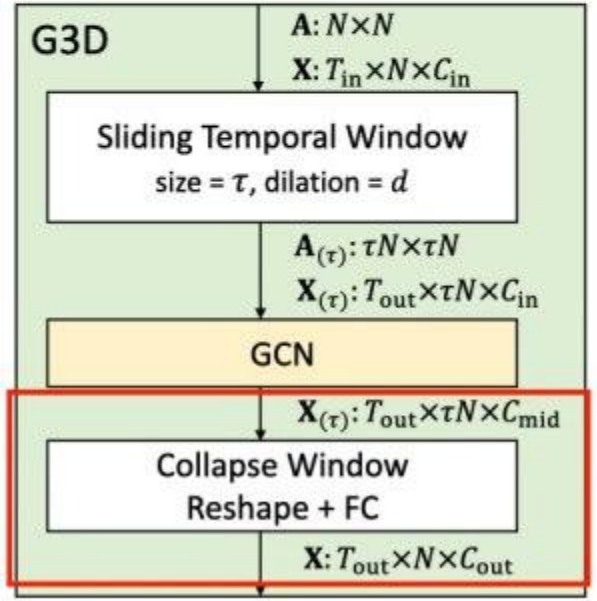




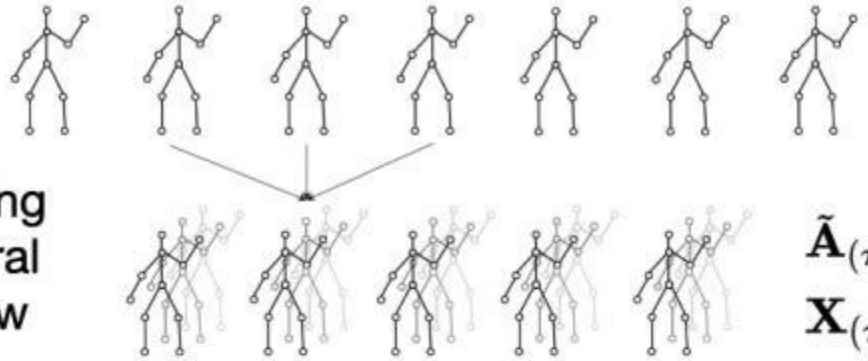
MS-G3D

- Unified spatial-temporal modeling





(1) Sliding Temporal Window



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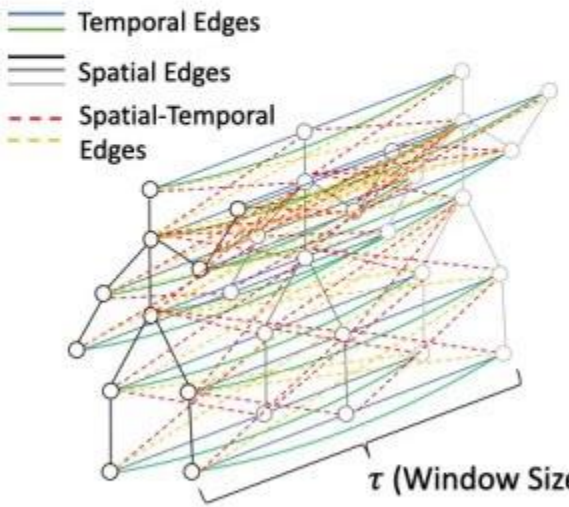

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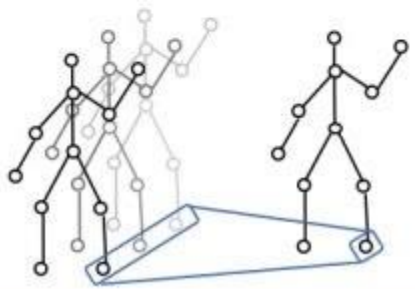
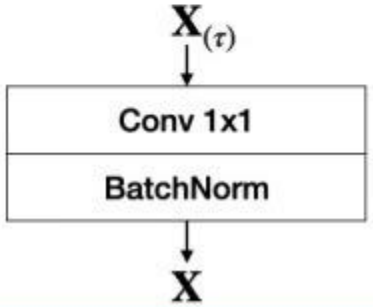
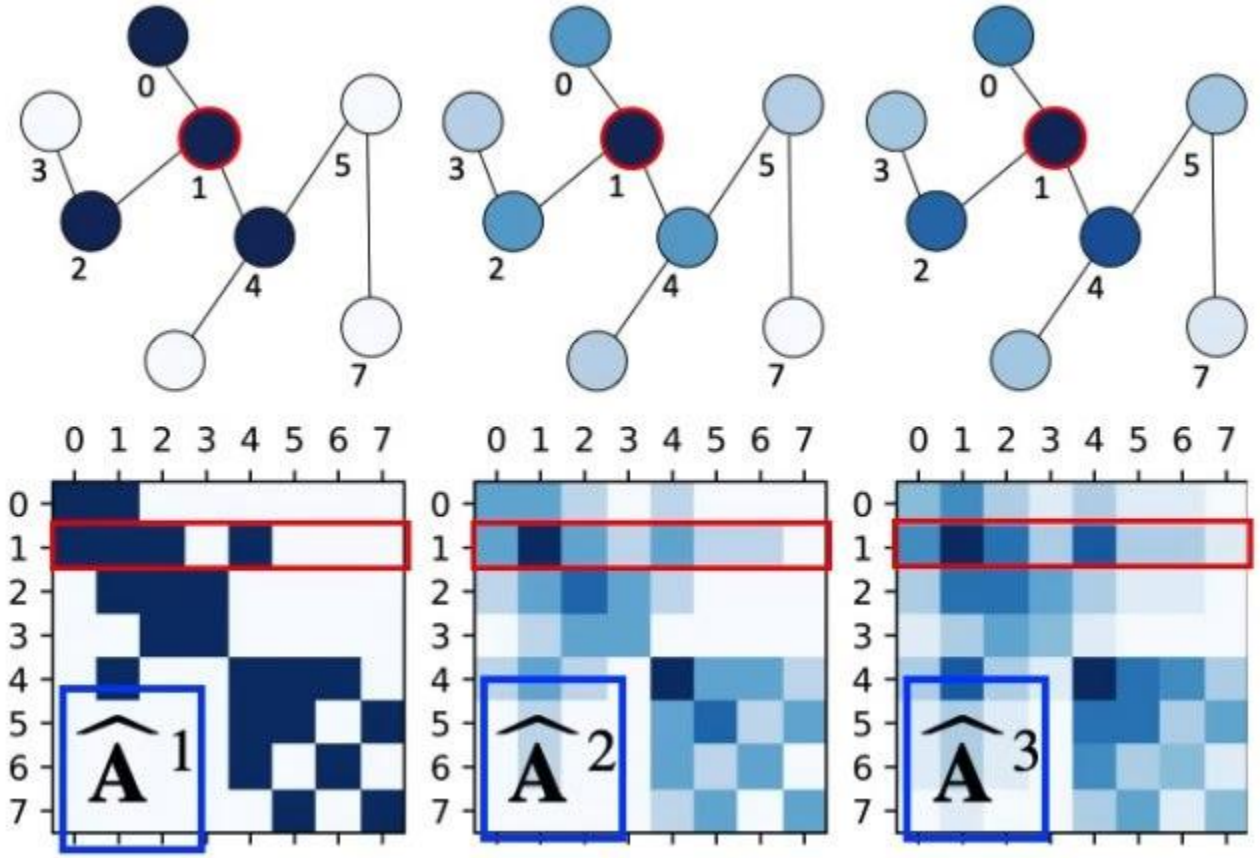




MS-G3D

- Disentangled Multi-scale Aggregation:
  - Making k-hop neighbors reachable



Multi-Scale Aggregation

with different $\hat{\mathbf{A}}^k$ and $\Theta_{(k)}$

$$\mathbf{X}_t^{(l+1)} = \sigma\left(\sum_{k=0}^{K} \hat{\mathbf{A}}^k \mathbf{X}_t^{(l)} \Theta_{(k)}^{(l)}\right)$$

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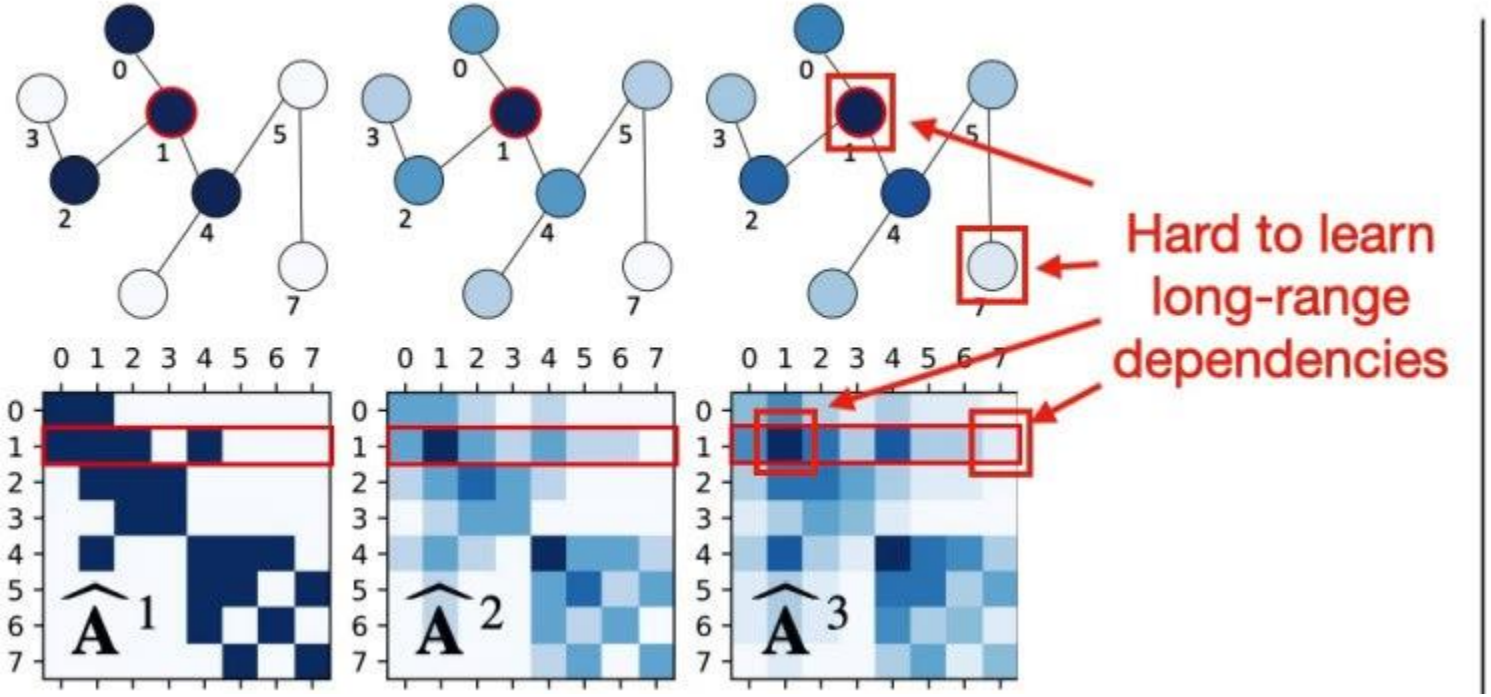
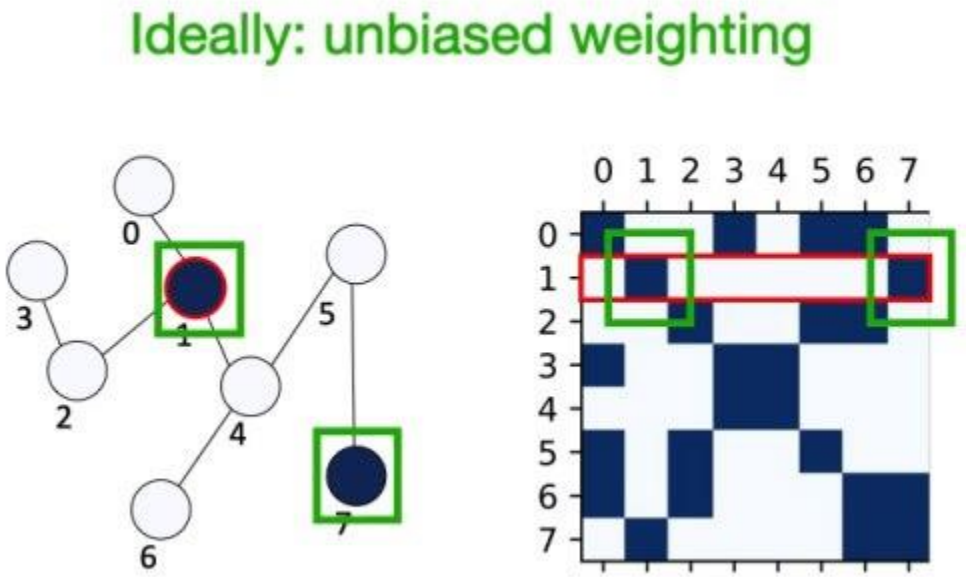
MS-G3D

- Disentangled Multi-scale Aggregation:
  - Biased node weighting:
    - Node weights are biased towards closer nodes
    - On skeleton graph, features from local body parts will dominate under GCNs





- Disentangled Multi-scale Aggregation:
  - AS-GCN employ higher-order polynomials of adjacency matrix to aggregate multi-scale structural information at time t

$$\mathbf{X}_t^{(l+1)} = \sigma\left(\sum_{k=0}^{K} \widehat{\mathbf{A}}^k \mathbf{X}_t^{(l)} \Theta_{(k)}^{(l)}\right)$$

  - However, it is more possible length k walks to closer nodes than actual k-hop neighbors due to cyclic walks
  - Disentangling neighborhoods:
    - Define k-adjacency matrix:

$$[\tilde{\mathbf{A}}_{(k)}]_{i,j} = \begin{cases} 1 & \text{if } d(v_i, v_j) = k, \\ 1 & \text{if } i = j, \\ 0 & \text{otherwise,} \end{cases}$$

: shortest distance in hops between 2 nodes

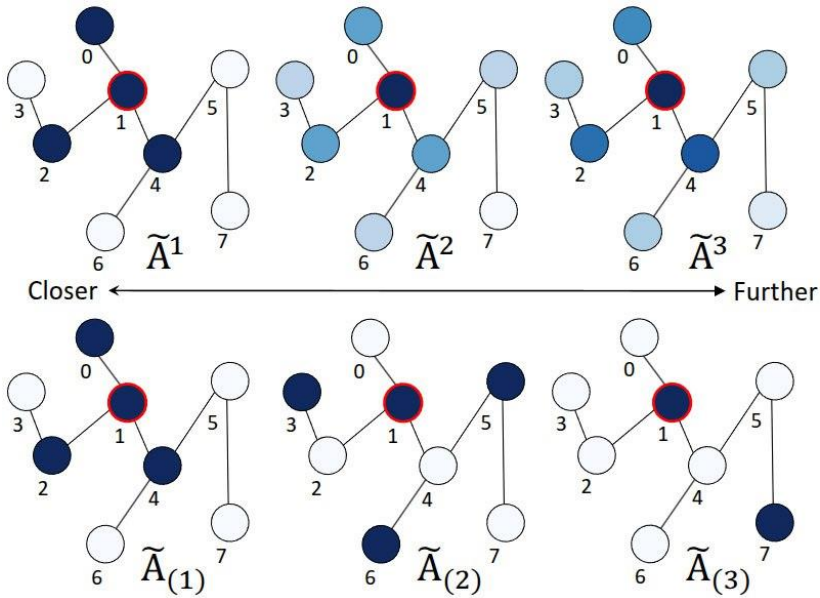


Figure : top: closer nodes receive higher weight, bottom: proposed aggregation models keep identity features

- Disentangled Multi-scale Aggregation:
  - Updated multi-scale graph convolutions:

$$\boxed{\widehat{\mathbf{A}} = \tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}}$$

$$\mathbf{X}_t^{(l+1)} = \sigma\left(\sum_{k=0}^{K} \boxed{\widehat{\mathbf{A}}^k}\mathbf{X}_t^{(l)}\Theta_{(k)}^{(l)}\right)$$

$$\longrightarrow$$

$$\boxed{\widehat{\mathbf{A}}_{(k)} = \tilde{\mathbf{D}}_{(k)}^{-\frac{1}{2}}\tilde{\mathbf{A}}_{(k)}\tilde{\mathbf{D}}_{(k)}^{-\frac{1}{2}}}$$

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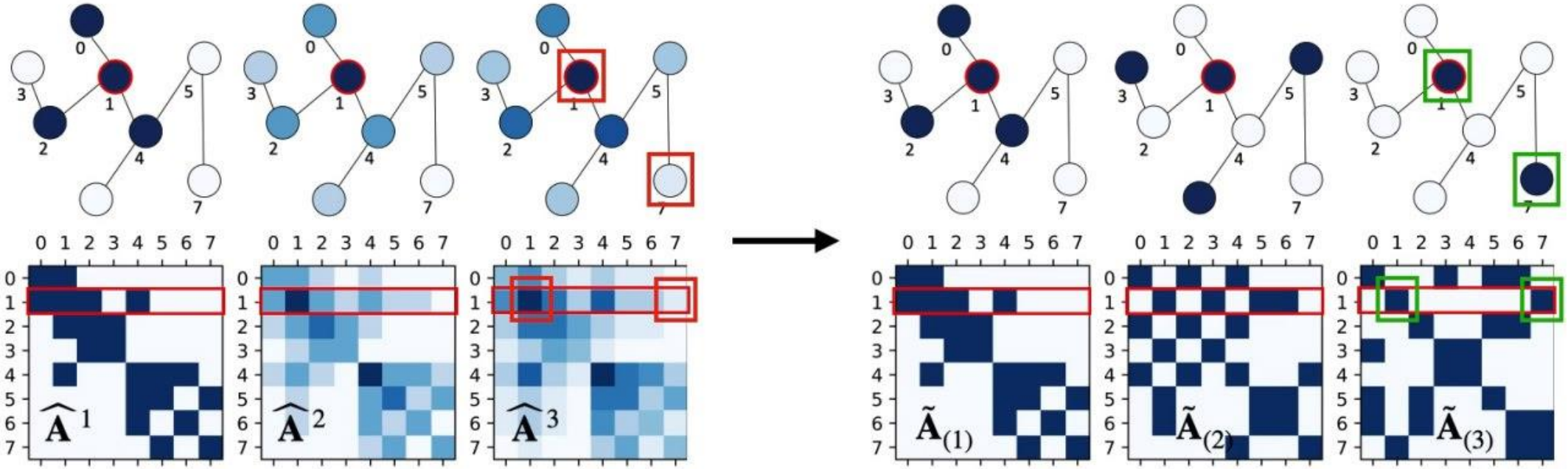

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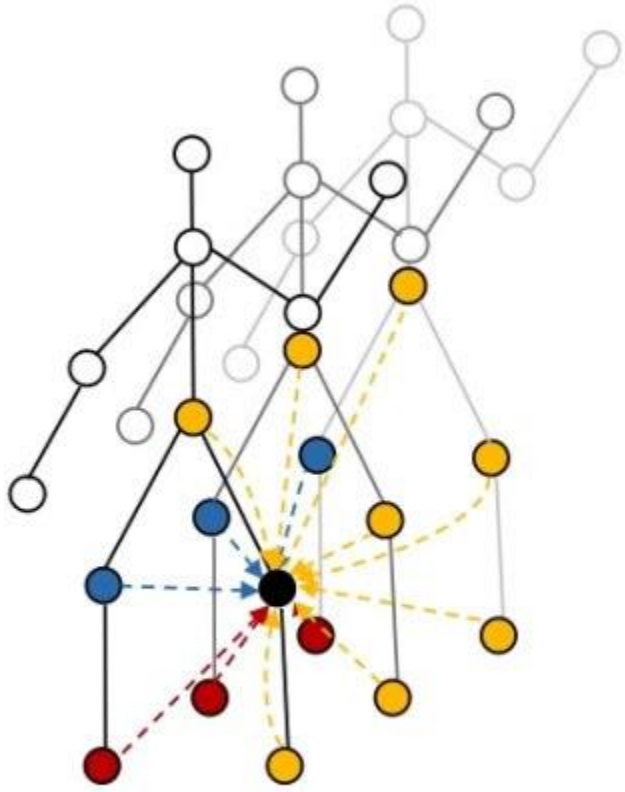
MS-G3D

- G3D: unified spatial-temporal modeling
  - Multi-scale learning directly over spatial-temporal domain
  - Disentangled aggregation complements G3D due to its higher node degrees



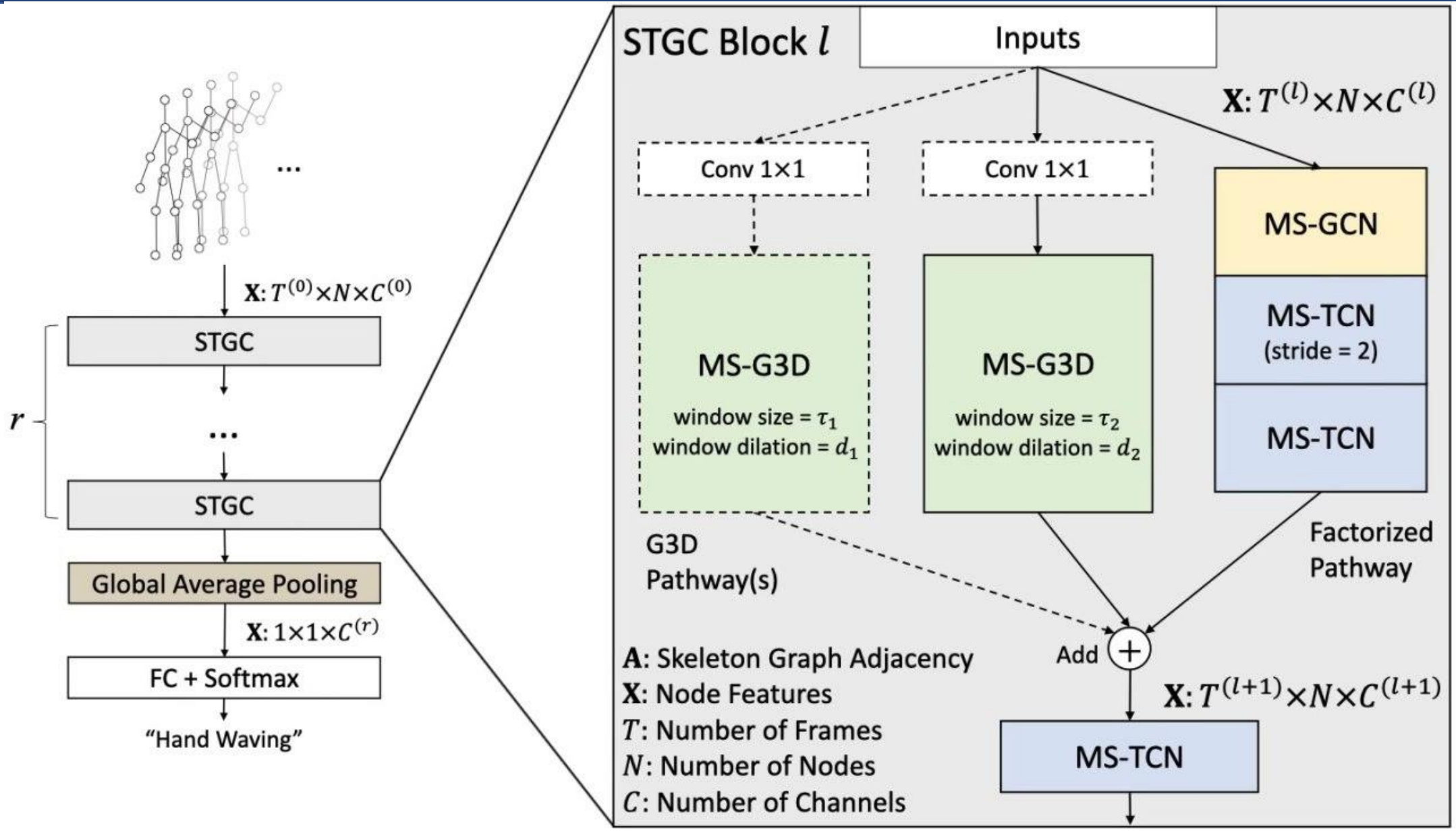
(partially coloured for visual clarity)

Unified Spatial-Temporal Modeling

$$\left[\mathbf{X}^{(l+1)}_{(\tau)}\right]_t = \sigma\left(\sum_{k=0}^{K} \tilde{\mathbf{D}}^{-\frac{1}{2}}_{(\tau,k)} \tilde{\mathbf{A}}_{(\tau,k)} \tilde{\mathbf{D}}^{-\frac{1}{2}}_{(\tau,k)} \left[\mathbf{X}^{(l)}_{(\tau)}\right]_t \Theta^{(l)}_{(k)}\right)$$

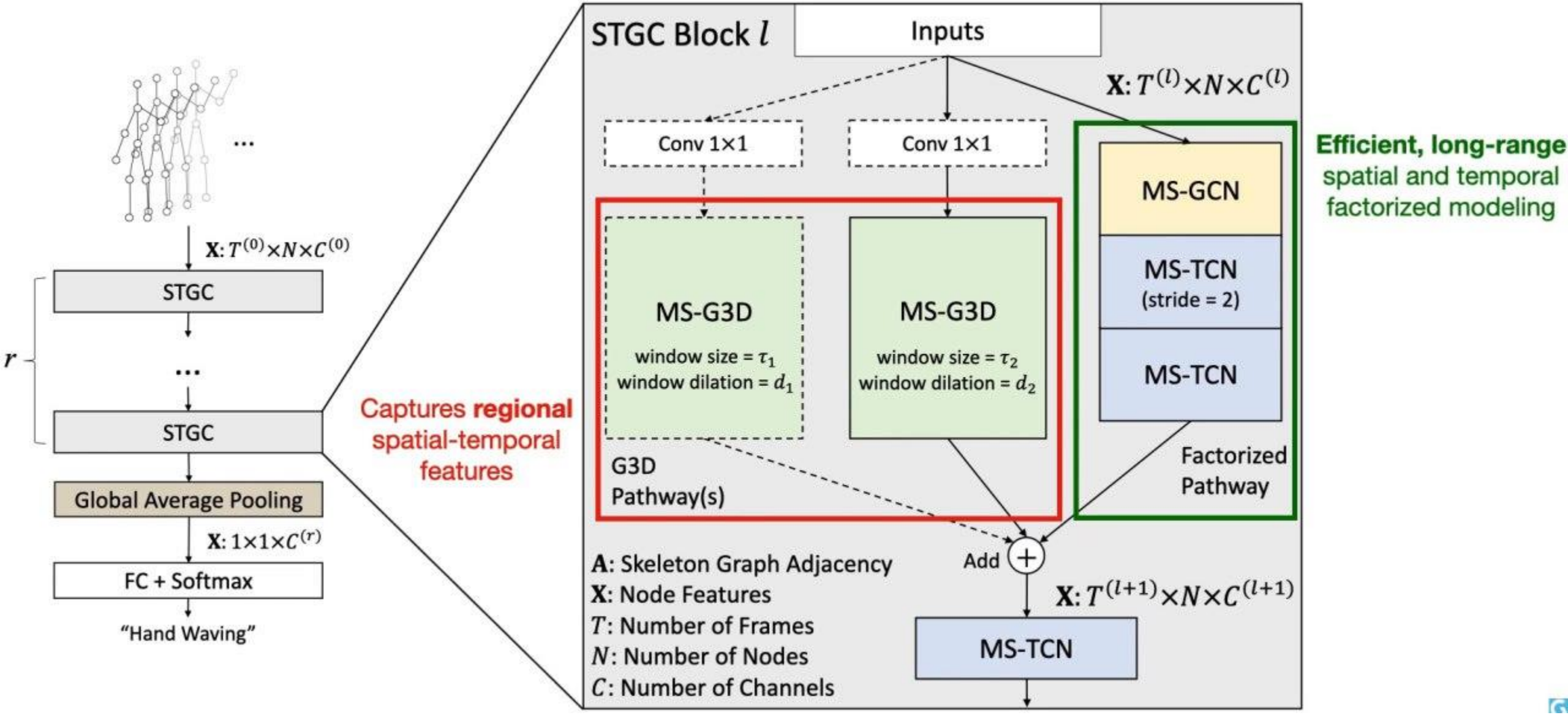
Disentangled Multi-Scale Aggregation

MS-G3D



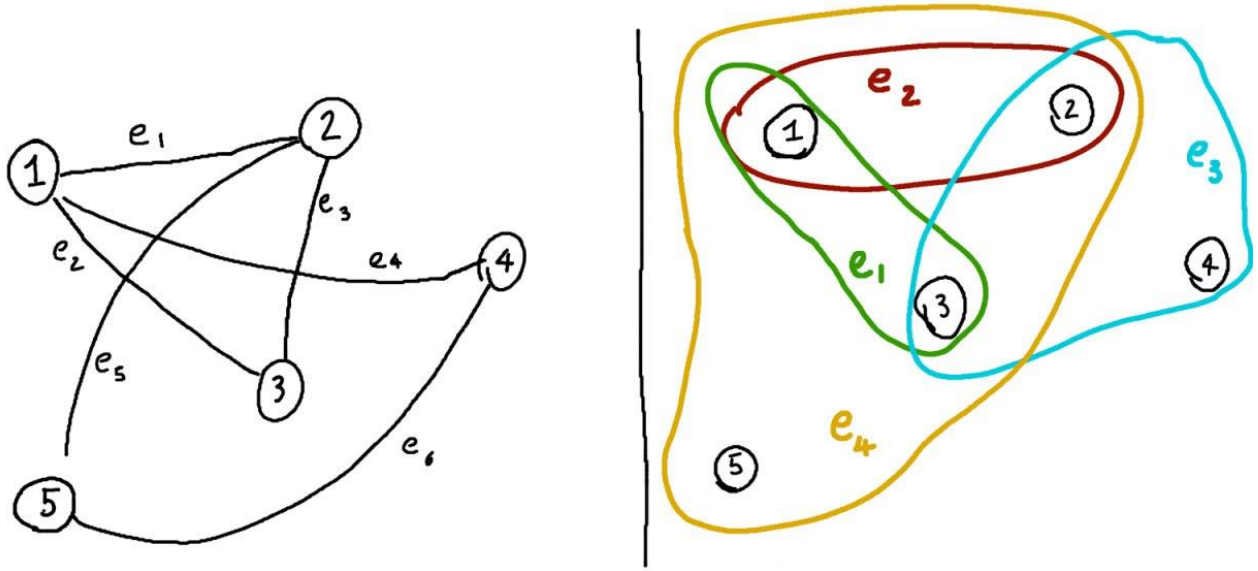
Disentangling and Unifying Graph Convolutions for Skeleton-Based Action Recognition, CVPR 2020

MS-G3D



Disentangling and Unifying Graph Convolutions for Skeleton-Based Action Recognition, CVPR 2020

- Why hypergraph?
  - Hyperedge connects more than 2 nodes
  - In previous works, we can see many motion involve higher-order relations



Graph Hypergraph

- Incidence matrix:
  - Nodes can belong to more than 2 hyperedges
  - Hypergraph G = (V, E, W)

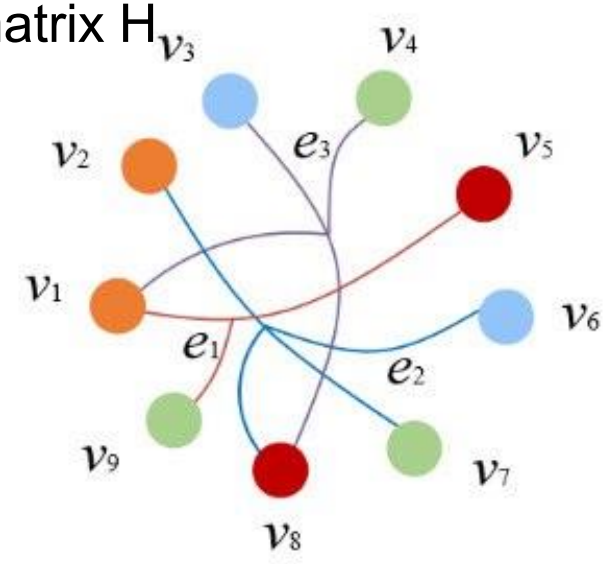
  Where:

  V is set of nodes

  E is set of hyperedges

  W is diagonal matrix
  - Hypergraph is represented by incidence matrix H

$$h(v, e) = \begin{cases} 1, & v \in e \\ 0, & v \notin e \end{cases}$$



$$\begin{array}{c|ccc} & e_1 & e_2 & e_3 \\ \hline v_1 & 1 & 0 & 1 \\ v_2 & 0 & 1 & 0 \\ v_3 & 0 & 0 & 1 \\ v_4 & 0 & 0 & 1 \\ v_5 & 1 & 0 & 0 \\ v_6 & 0 & 1 & 0 \\ v_7 & 0 & 1 & 0 \\ v_8 & 0 & 1 & 1 \\ v_9 & 1 & 0 & 0 \end{array}$$

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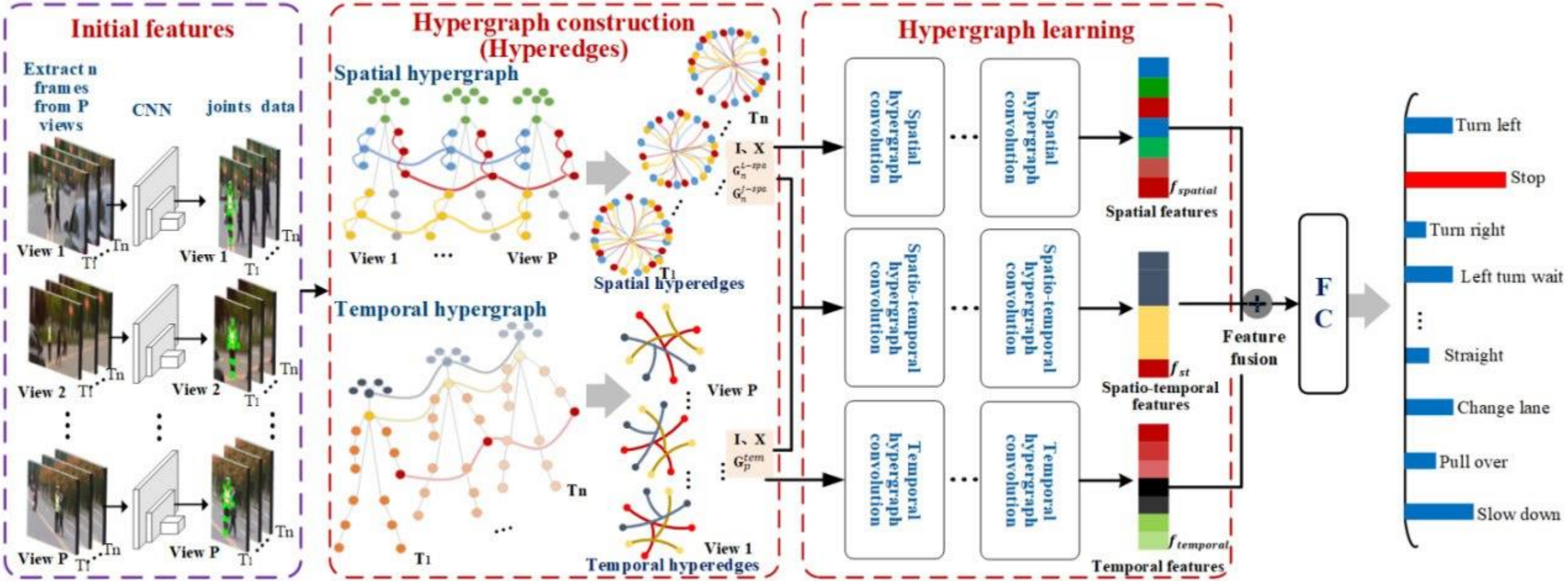
DG-STGCN

- Overall architecture:



Multi-View Time-Series Hypergraph Neural Network for Action Recognition, IEEE Transactions on Image Processing 2024

DG-STGCN

- Multi-view hypergraph construction
  - Multi-view spatial hypergraph construction:
    - 2 construction strategies based on limb components and adjacent joints
- Limb construction strategy (LCSH)

$$\mathcal{G}^{\text{L-}spa} = (\mathcal{V}^{\text{L-}spa}, \mathcal{E}^{\text{L-}spa}, \mathbf{W}^{\text{L-}spa})$$

$$\mathcal{V}^{\text{L-}spa} = \{\mathcal{V}_1^{\text{L-}spa}, \mathcal{V}_2^{\text{L-}spa}, \cdots, \mathcal{V}_m^{\text{L-}spa}\}$$

$$\mathcal{E}^{\text{L-}spa} = \{e_1^{\text{L-}spa}, e_2^{\text{L-}spa}, \cdots, e_m^{\text{L-}spa}\}$$

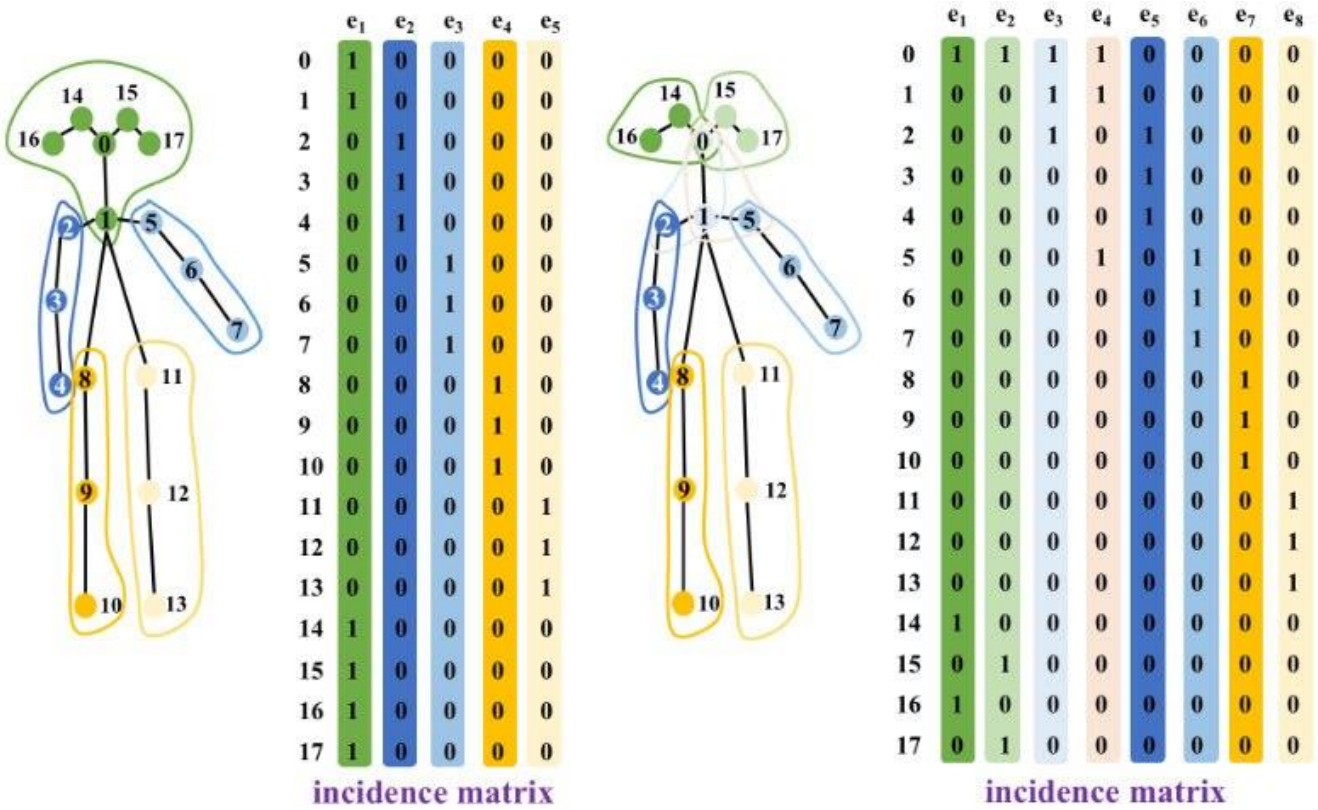
$\mathbf{W}^{\text{L-}spa}$: weight matrix

  - 5 parts: trunk, left/right hand, left/right leg
- Incidence matrix of spatial hypergraph

$$\mathbf{H}_n^{\text{L-}spa}(v_{p,n}^{(i)}, e_{m,n}^{\text{L-}spa}) = \begin{cases} 1 & v_{p,n}^{(i)} \in e_{m,n}^{\text{L-}spa} \\ 0 & v_{p,n}^{(i)} \notin e_{m,n}^{\text{L-}spa} \end{cases}$$

- Laplace matrix:

$$\mathbf{G}_n^{\text{L-}spa} = (\mathbf{D}_{v_n}^{\text{L-}spa})^{-\frac{1}{2}} \mathbf{H}_n^{\text{L-}spa} \mathbf{W}_n^{\text{L-}spa} (\mathbf{D}_{e_n}^{\text{L-}spa})^{-1} (\mathbf{H}_n^{\text{L-}spa})^{\mathrm{T}} (\mathbf{D}_{v_n}^{\text{L-}spa})^{-\frac{1}{2}}$$



(a) Limb construction strategy (b) Adjacent joints construction strategy

DG-STGCN

- Multi-view hypergraph construction
  - Multi-view spatial hypergraph construction:
    - 2 construction strategies based on limb components and adjacent joints
- Adjacent joints construction strategy (AJSH)
  - Composed by 3 adjacent joints
- Laplace matrix:

$$\mathbf{G}_n^{\text{J-}spa} = (\mathbf{D}_{v_n}^{\text{J-}spa})^{-\frac{1}{2}} \mathbf{H}_n^{\text{J-}spa} \mathbf{W}_n^{\text{J-}spa} (\mathbf{D}_{e_n}^{\text{J-}spa})^{-1} (\mathbf{H}_n^{\text{J-}spa})^{\mathrm{T}} (\mathbf{D}_{v_n}^{\text{L-}spa})^{-\frac{1}{2}}$$

(a) Limb construction strategy (b) Adjacent joints construction strategy

DG-STGCN

- Multi-view hypergraph construction
  - Multi-view temporal hypergraph construction:
    - Divide sequential frames into a group from same view

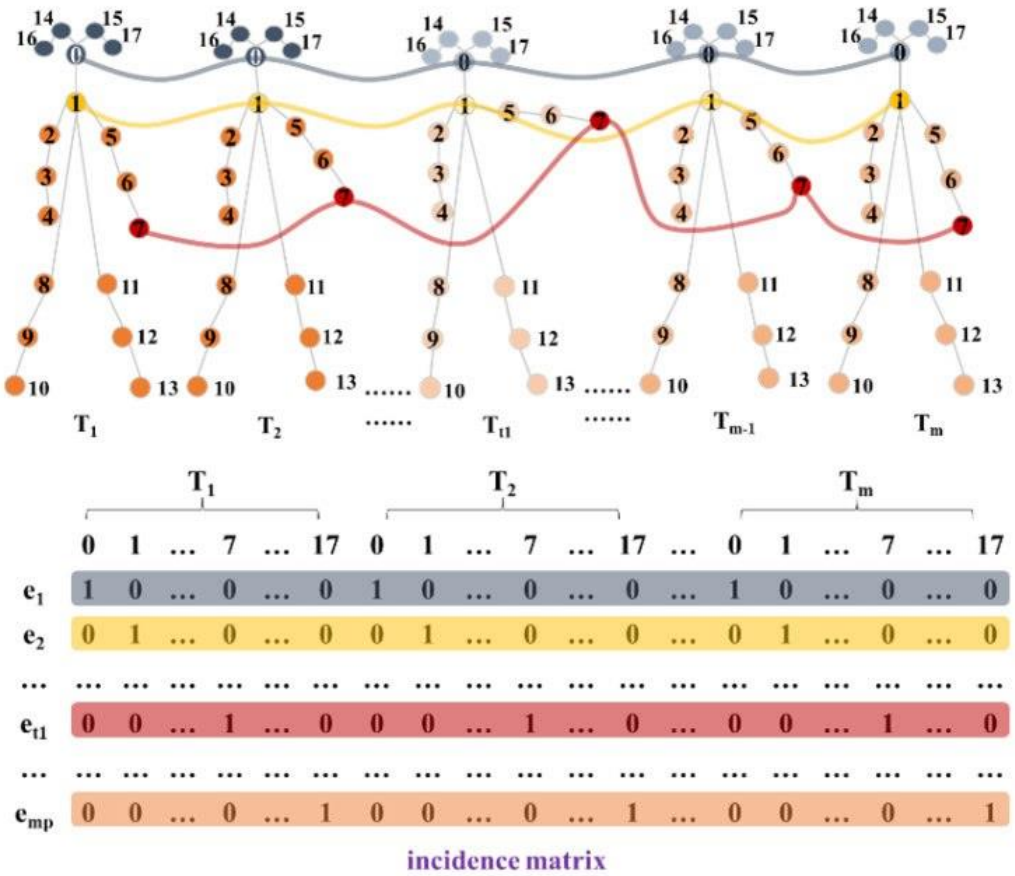
$$\mathcal{G}^{tem} = (\mathcal{V}^{tem}, \mathcal{E}^{tem}, \mathbf{W}^{tem}).$$

    - There are P views, so initial incidence matrix of each temporal hypergraph:

$$\mathbf{H}_p^{tem}(v_{p,n}^{(i)}, e_{m,p}^{tem}) = \begin{cases} 1 & v_{p,n}^{(i)} \in e_{m,p}^{tem} \\ 0 & v_{p,n}^{(i)} \notin e_{m,p}^{tem} \end{cases}$$

    - Laplace matrix:

$$\mathbf{G}_p^{tem} = \mathbf{D}_{v_p}^{-\frac{1}{2}} \mathbf{H}_p^{tem} \mathbf{W}_p^{tem} \mathbf{D}_{e_p}^{-1} (\mathbf{H}_p^{tem})^{\mathrm{T}} \mathbf{D}_{v_p}^{-\frac{1}{2}}$$



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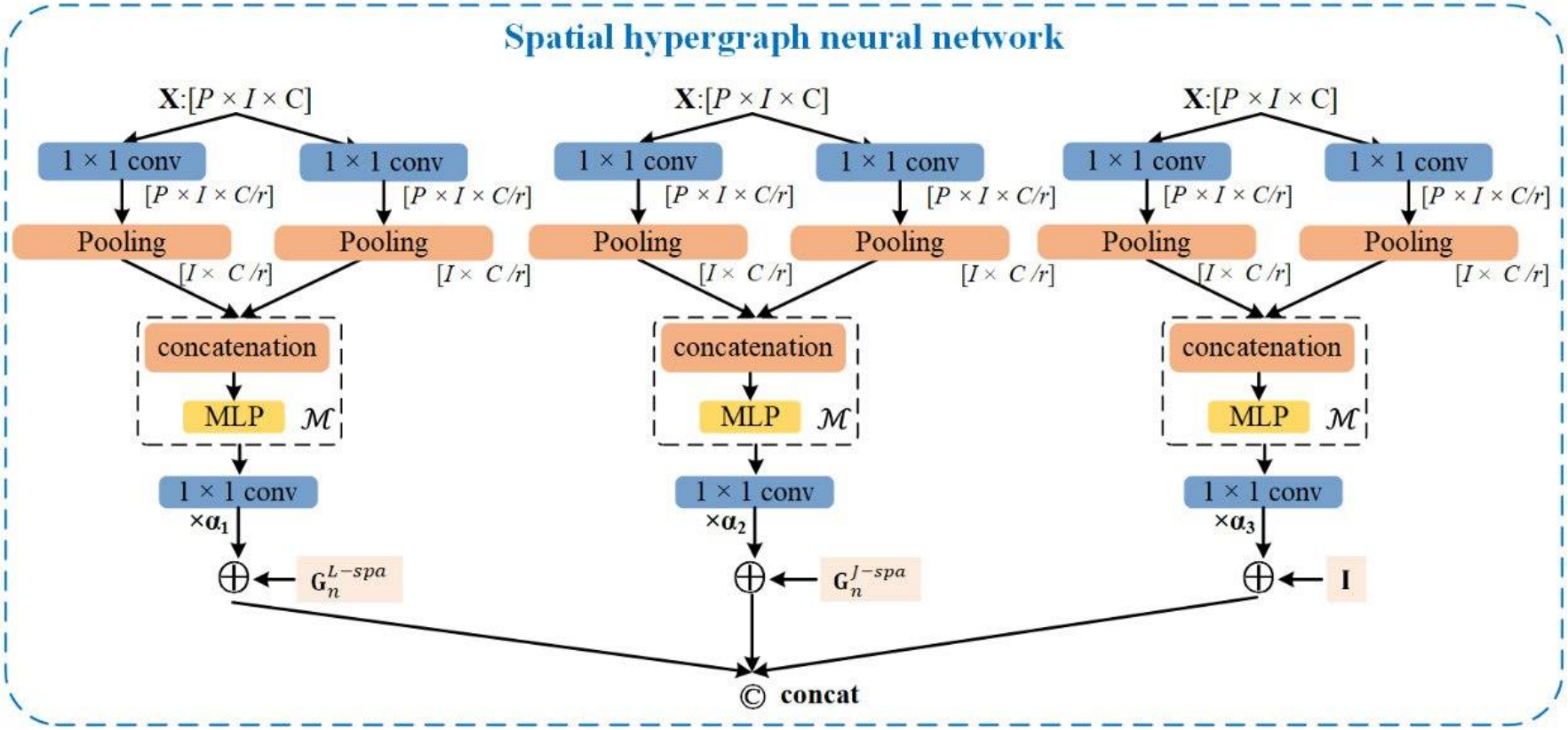
- Multi-view time-series hypergraph learning:
  - Spatial hypergraph neural network

$$f_{spatial} = [(\mathbf{G}_n^{\text{L-}spa} + \alpha_1 \cdot \mathcal{M}(\mathbf{X}))\mathbf{X}\mathbf{W}\| (\mathbf{G}_n^{\text{J-}spa} + \alpha_2 \cdot \mathcal{M}(\mathbf{X}))\mathbf{X}\mathbf{W}\| (\mathbf{I} + \alpha_3 \cdot \mathcal{M}(\mathbf{X}))\mathbf{X}\mathbf{W}]$$

$$\mathcal{M}(\mathbf{X}) = MLP\left(\psi(x_{1,1}^{(1)})\| \cdots \|\psi(x_{p,n}^{(i)})\right)$$



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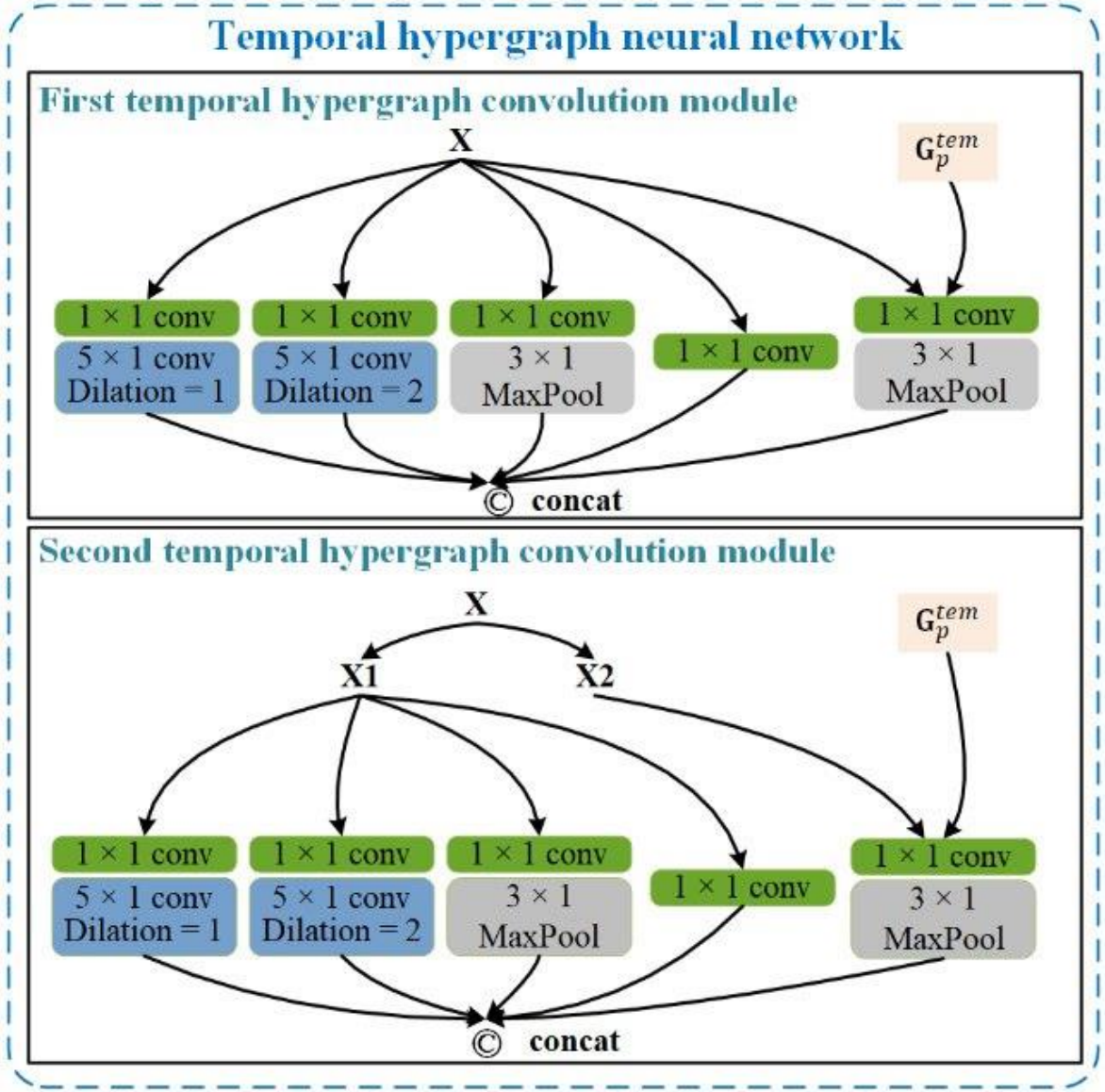
- Multi-view time-series hypergraph learning:
  - Temporal hypergraph neural network
    - First convolution module capture global dependencies among sequence frames
    - Second convolution moule divide node features into X1, X2, to efficiently learn frames in different time lengths and action diversity of actions

$$
\begin{aligned}
f_1 = \text{concat}\,(&\text{conv}_{d=1}(\text{conv}(X)) \\
&+ \text{conv}_{d=2}(\text{conv}(X)) \\
&+ \text{MaxPool}(\text{conv}(X)) + \text{conv}(X) \\
&+ \text{MaxPool}\left(\text{conv}(X) + \mathbf{G}_p^{\text{tem}}\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
f_2 = \text{concat}\,(&\text{conv}_{d=1}\,(\text{conv}\,(X_1)) \\
&+ \text{conv}_{d=2}\,(\text{conv}\,(X_1)) \\
&+ \text{MaxPool}\,(\text{conv}\,(X_1)) \\
&+ \text{conv}\,(X_1) \\
&+ \text{MaxPool}\left(\text{conv}\,(X_2) + \mathbf{G}_p^{\text{tem}}\right)\Big)
\end{aligned}
$$



Multi-View Time-Series Hypergraph Neural Network for Action Recognition, IEEE Transactions on Image Process

DG-STGCN

- Multi-view time-series hypergraph learning:
  - Spatio-temporal hypergraph neural network
    - Learn spatio-temporal features of action sequences
    - Input:
      - Node spatial features
      - Node temporal features
      - (Spatial)Limb hypergraph
      - (Spatial)Adjacent hypergraph
      - Temporal hypergraph
      - Total number of nodes
    - Output: f_{st}

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Comparison

Method	Graph type	Topology	Temporal modeling	Core
ST-GCN	Graph	Fixed physical	1D temporal conv	Neighbor aggregation
2s-AGCN	Graph	Adaptive	Temporal conv per layer	2 streams (joint & bone)
AS-GCN	Graph	Actional + structural	Temporal conv + future pose head	Higher-order
MS-G3D	Graph	Multiscale, disentangled	G3D unified ST conv + tempo ral conv	Dense cross-spacetime edges
DG-STGCN	Hypergraph	Multiview hyperedge	Time-series hypergraph layers	Higher-order, multi-view modeling

Q
A